

6.12.2023

Unit - 2

EM

classmate

Date

Page

Beam :- Beam are the horizontal structural in which length is larger than other two dimension.

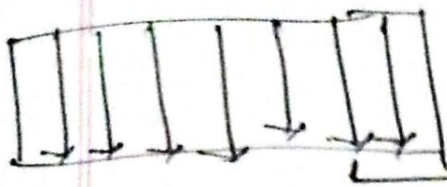
Load :- There are basically three types of load -

- Point load
- Uniformly distributed load
- Uniformly varying load

Types of Support → There are 4 types of support.

- fixed support
- hinge / pin support
- Roller support
- simple support

Types of Beam → There are basically 4 types of beam.



classmate

Date _____

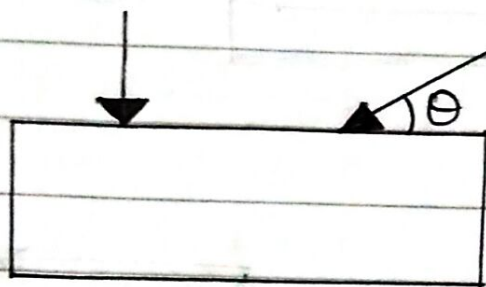
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UDL $\Rightarrow$ Area of Triangle $\cdot L \times b$

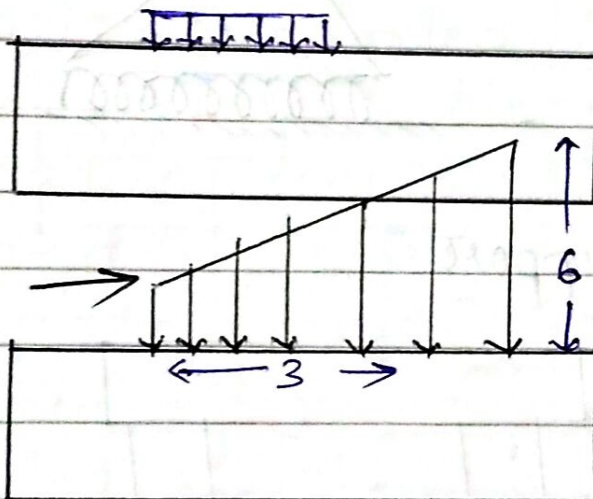
$$\Rightarrow \frac{b}{2}$$

- $\rightarrow$ Simple supported beam.
- $\rightarrow$ over hanging beam.
- $\rightarrow$ cantilever beam.
- $\rightarrow$ fixed beam

load-



Point load-



Uniformly load

$$\text{Total load} \Rightarrow \frac{1}{2} \times b \times h$$

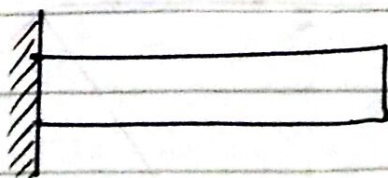
UVL $\rightarrow$

Uniformly varying load $\rightarrow$

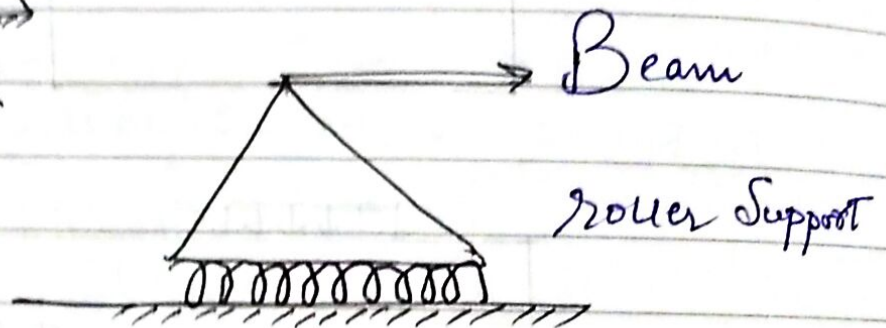
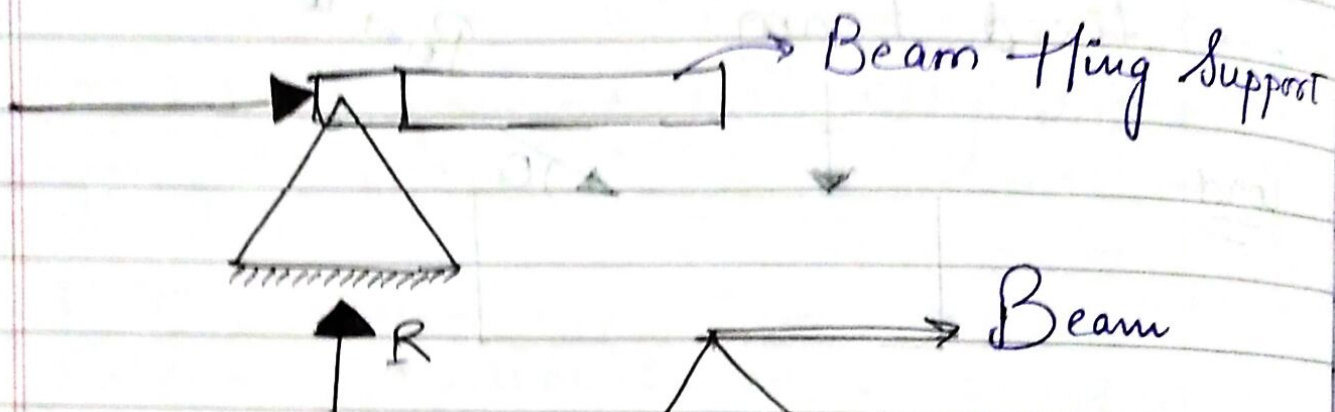
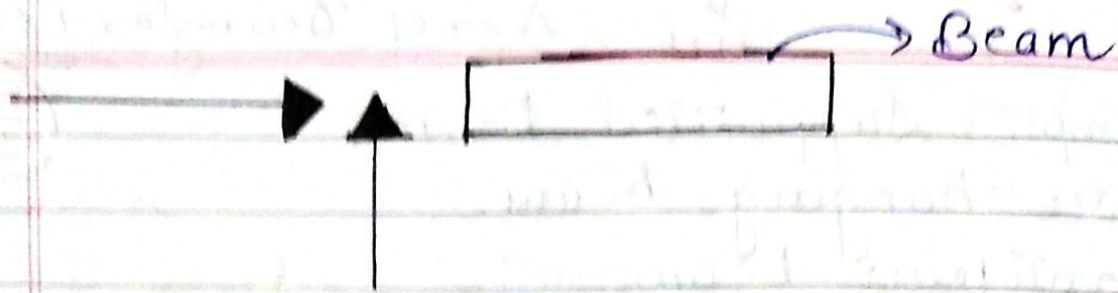
$$\frac{b}{3} \Rightarrow \frac{3}{3} \Rightarrow 1$$

Point load $\Rightarrow (\bar{x}-1)$ unit

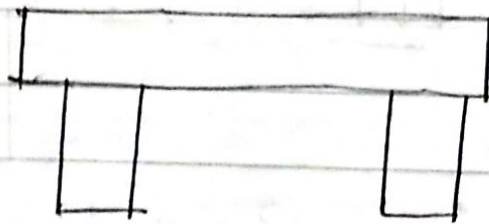
Support-



- fixed support

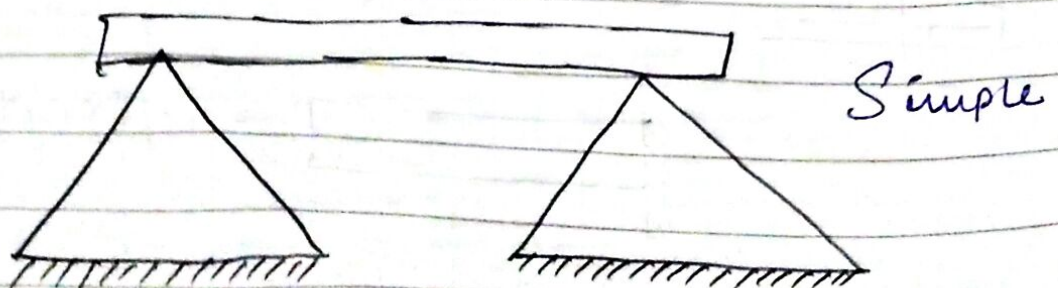


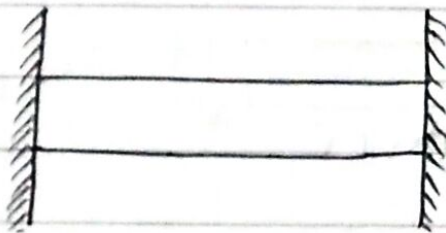
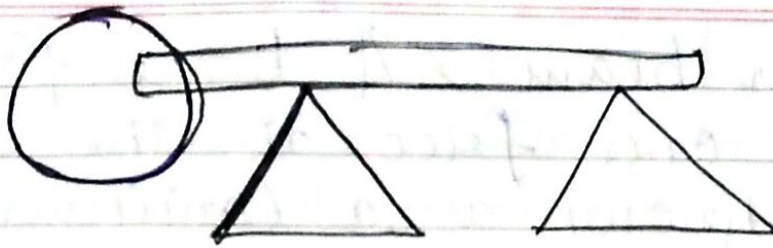
Simple Support -



Beam

→ Simple Support Beam -

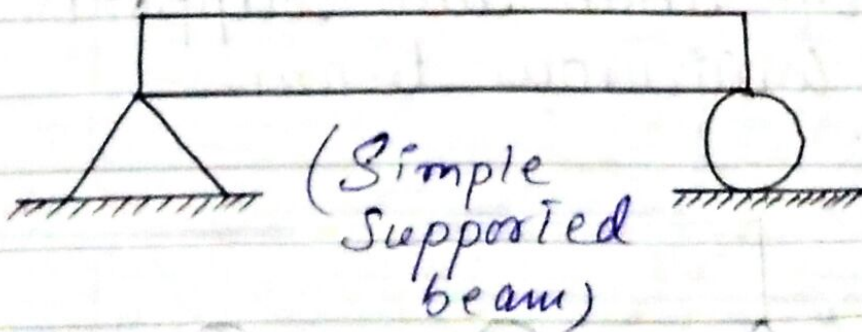




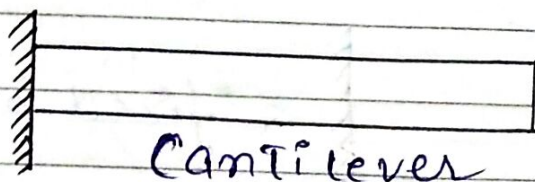
- fixed beam

Defⁿ

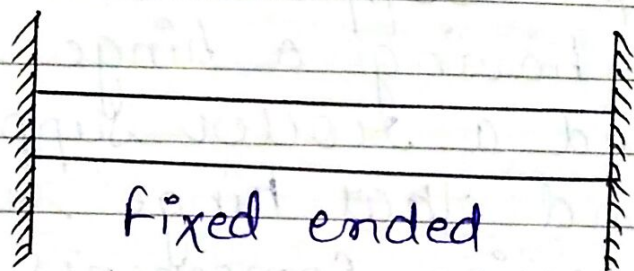
① Simple Supported beam $\rightarrow$ The beam having a hinge support at one end and a roller support at other end and that hinge support can have two reaction components, vertical and horizontal is known as simply supported beam.



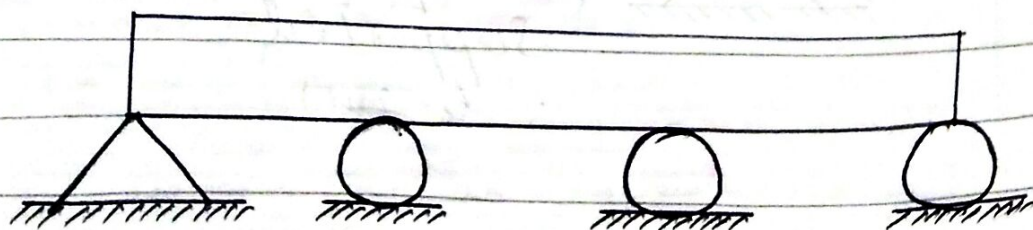
- ② Cantilever beam $\rightarrow$ A beam fixed at one end and free at the other end is known as a Cantilever beam.



- ③ Fixed beam :- A beam whose both ends are fixed, is known as a fixed beam.

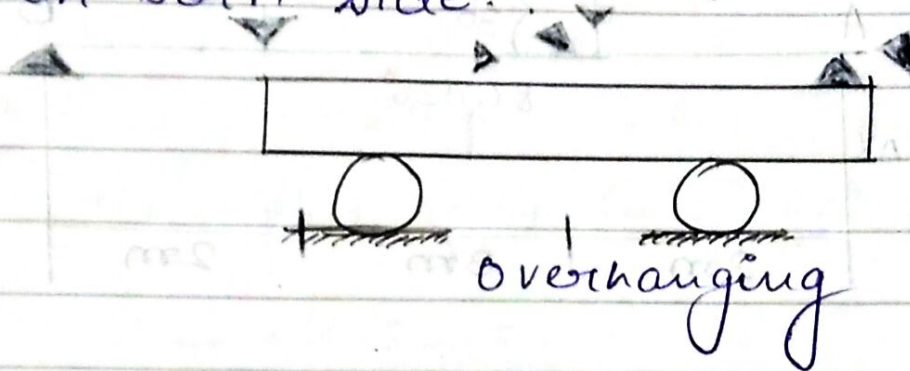


- ④ Continuous beam :- A beam supported on more than two supports is known as a continuous beam.

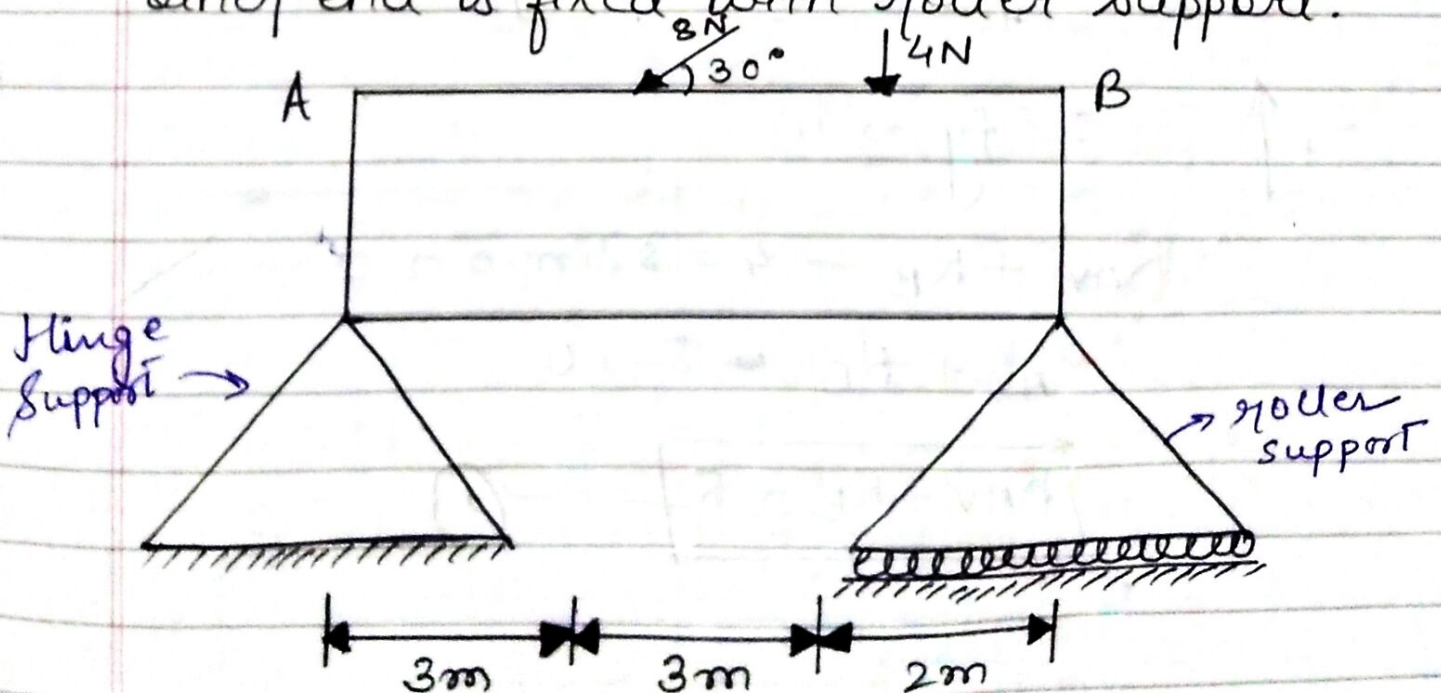


Continuous

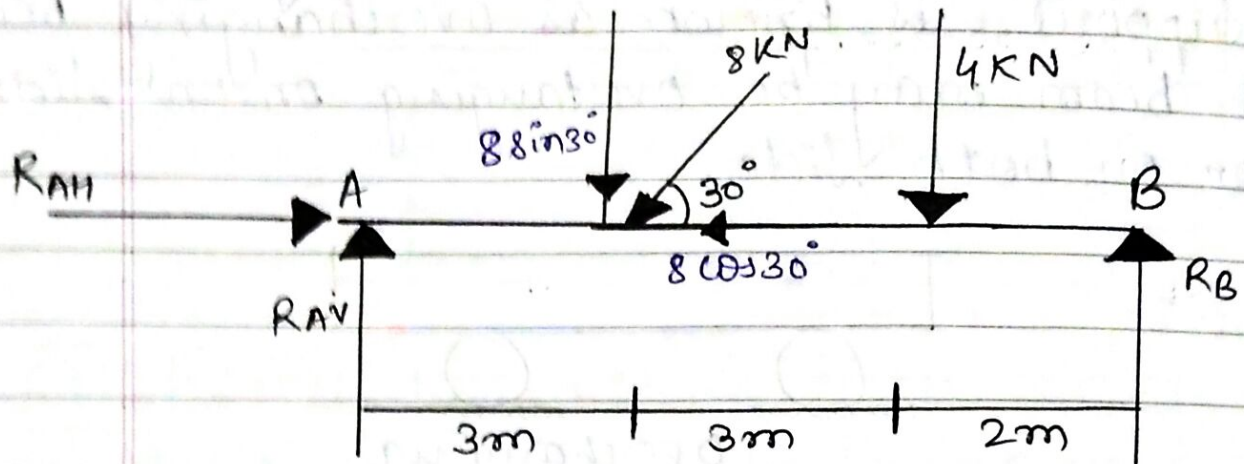
- ⑤ Overhanging beam :- A beam having its end portion extended beyond the support, is known as overhanging beam. A beam may be overhanging on one side or on both side.



- ⑧ Find the support and reaction of a beam AB as shown in the figure in which one end is fixed with hinge support and other end is fixed with roller support.



Sol FBD of Beam AB :-



$$\sum F_x \Rightarrow 0$$

$$R_{AH} - 8 \cos 30^\circ \Rightarrow 0$$

$$R_{AH} \Rightarrow 8 \times \cos 30^\circ \Rightarrow 6.928$$

$$\boxed{R_{AH} \Rightarrow 6.928 \text{ kN}}$$

$$\sum F_y \Rightarrow 0$$

$$R_{AV} + R_B - 4 - 8 \sin 30^\circ \Rightarrow 0$$

$$R_{AV} + R_B \Rightarrow 8 \Rightarrow 0$$

$$\boxed{R_{AV} + R_B \Rightarrow 8} \quad (1)$$

$$\sum M_A \rightarrow 0 \quad (\text{Moment at point A})$$

↳ Use point B
Incoming or outgoing
done ka force $\rightarrow 0$

$$\Rightarrow (8 \sin 30 \times 3 + 4 \times 6) - R_B \times 8 \Rightarrow 0$$

$$\Rightarrow 36 - R_B \times 8 \Rightarrow 0$$

$$R_B \Rightarrow \frac{36}{8} \Rightarrow \boxed{4.5 \text{ kN}}$$

Put the value of R_B in eq (1)

$$R_{AV} + 4.5 \Rightarrow 8$$

$$\boxed{R_{AV} \Rightarrow 3.5 \text{ kN}}$$

$$\text{Reaction at A} \Rightarrow \sqrt{(R_{AH})^2 + (R_{AV})^2}$$

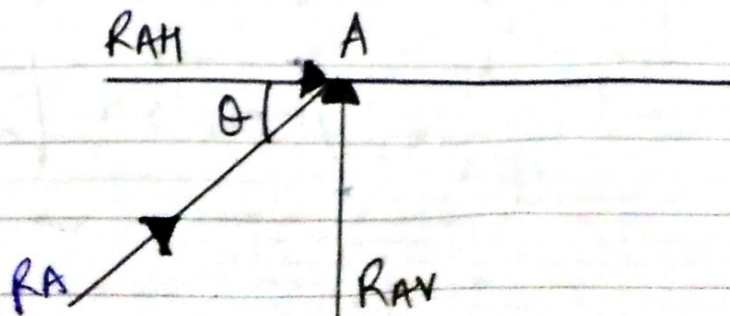
$$\Rightarrow \sqrt{(6.928)^2 + (3.5)^2}$$

$\Rightarrow \checkmark$

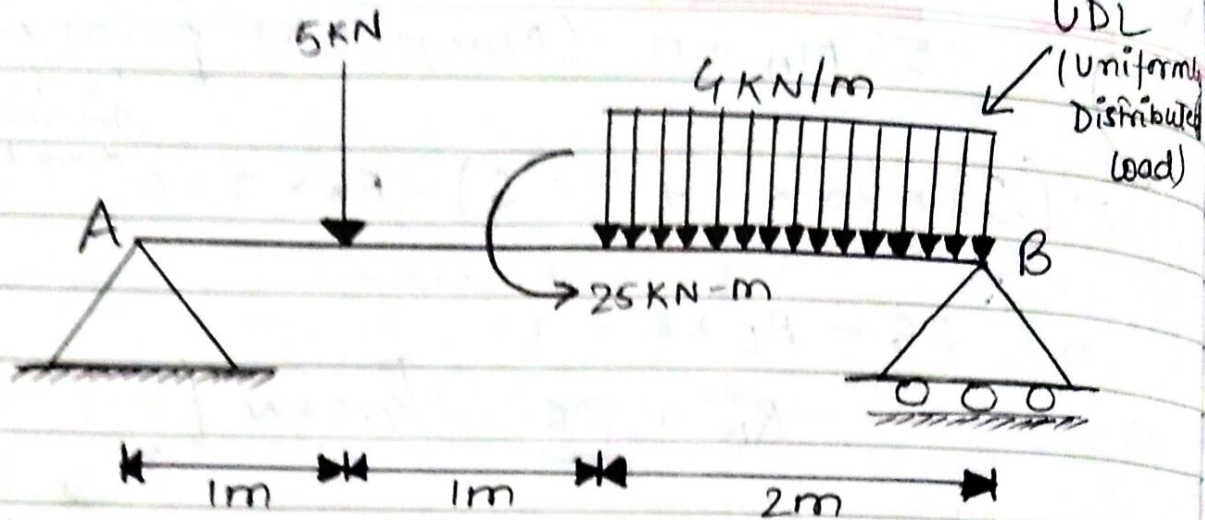
$$\boxed{\text{Reaction at A} \Rightarrow 7.761 \text{ kN}}$$

$$\theta \Rightarrow \tan^{-1} \left| \frac{3.5}{6.928} \right|$$

$$\boxed{\theta \Rightarrow 26.802^\circ}$$



Q



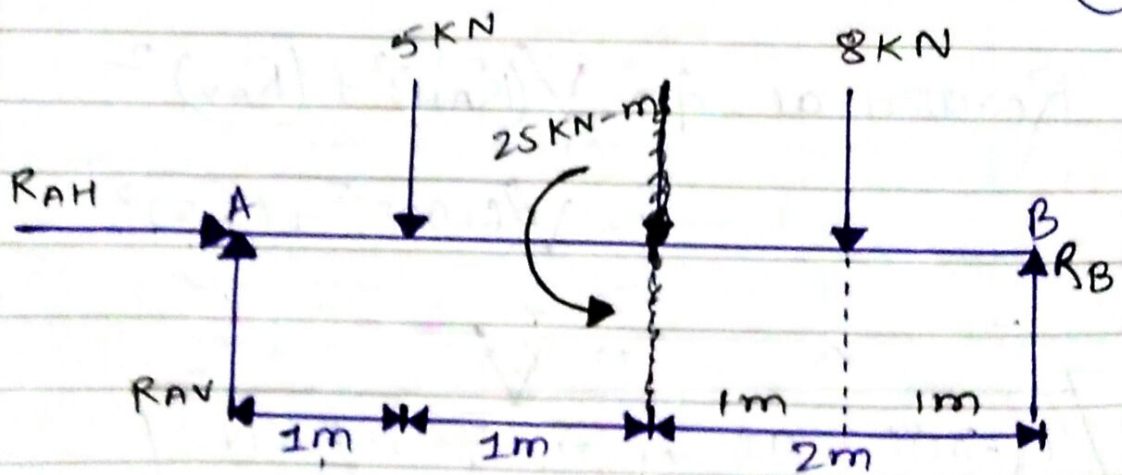
Find reaction & support?

$$UDL = 4 \times 2$$

$$\Rightarrow 8 \text{ kN}$$

$$= \left(\frac{6}{2} \right) \times \frac{2}{2} = 1 \text{ m}$$

Sol: FBD of beam AB :-



$$\rightarrow \sum f_x \Rightarrow 0$$

$$R_{AH} = 0$$



$$\sum f_y \Rightarrow 0$$

$$R_{AV} + R_B - 5 - 8 \Rightarrow 0$$

$$\boxed{R_{AV} + R_B \Rightarrow 13}$$



$$\sum M_A \Rightarrow 0$$

$$5 \times 1 + 8 \times 3 - 25 - R_B \times 4 \Rightarrow 0$$

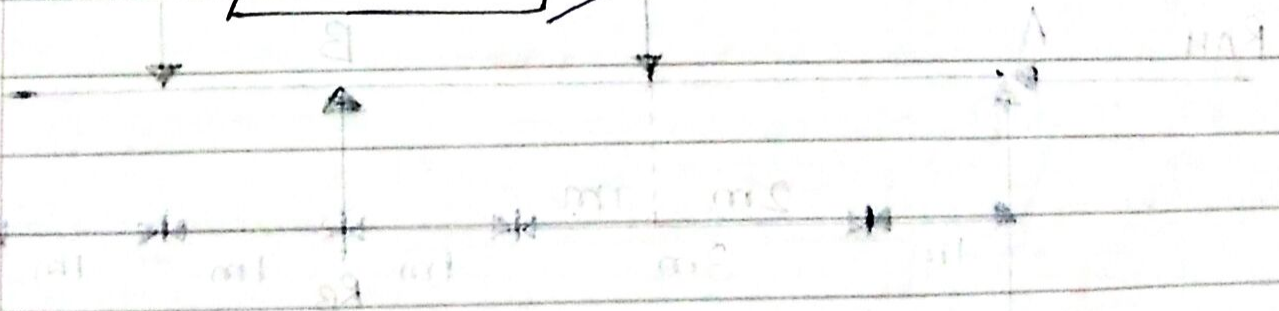
$$4 - R_B \times 4 \Rightarrow 0$$

$$\boxed{R_B \Rightarrow 1 \text{ kN}}$$

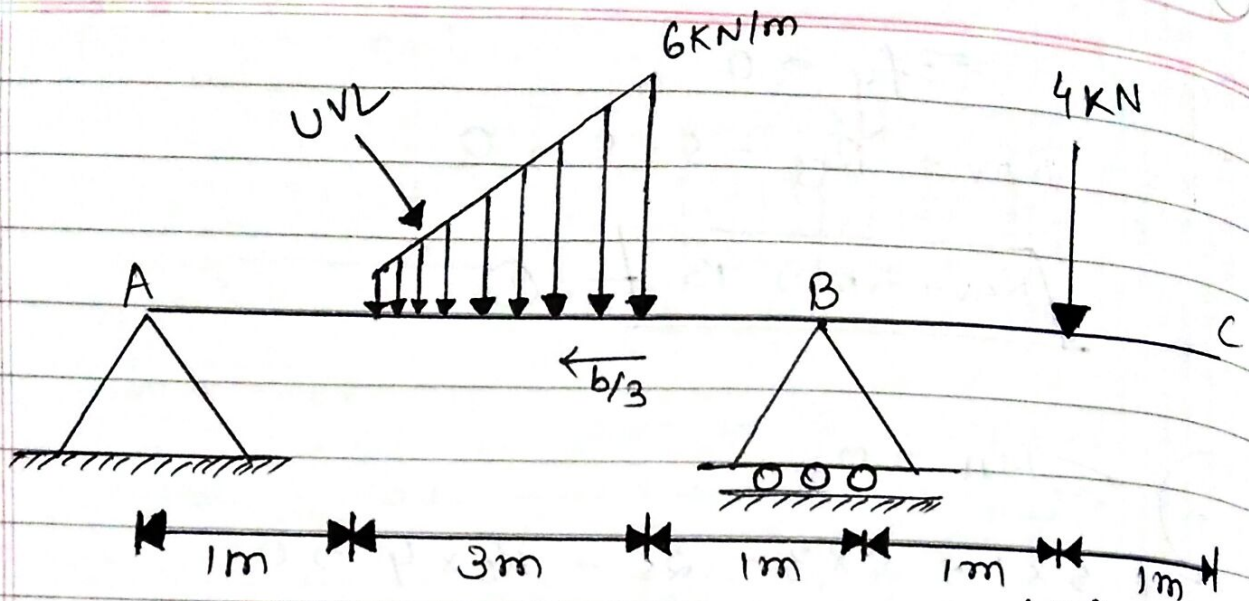
from (A) $\Rightarrow R_{AV} + 1 - 13 \Rightarrow 0$

$$R_{AV} - 12 \Rightarrow 0$$

$$\boxed{R_{AV} \Rightarrow 12 \text{ kN}}$$



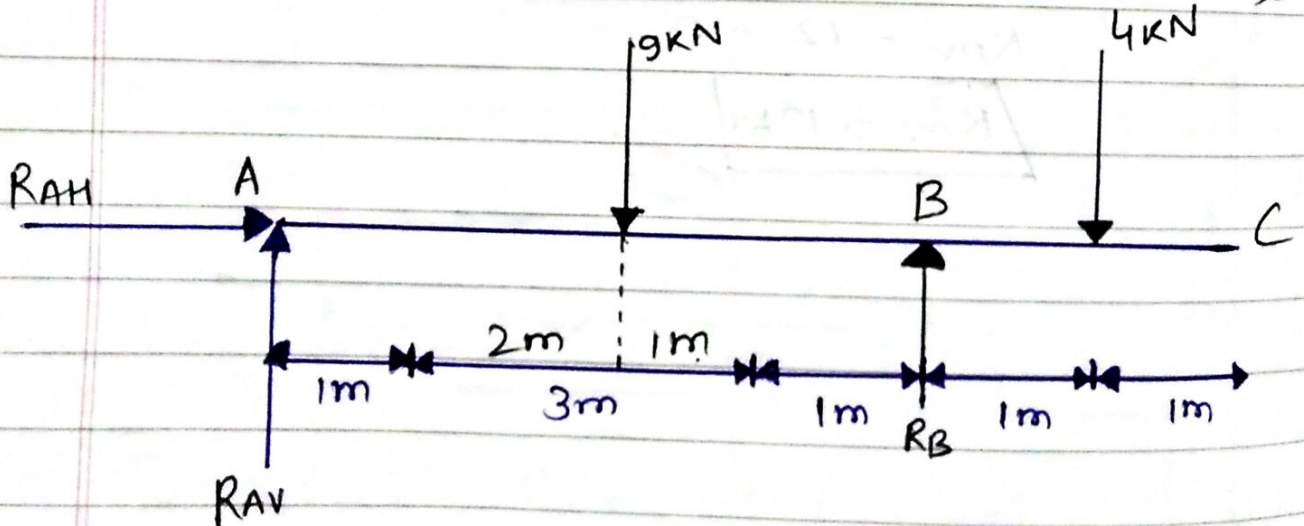
Q



find reaction and support at A & B.

Sol:FBD of Beam

$$\begin{aligned} \text{UDL} &\Rightarrow \frac{1}{2} \times 3 \times 6^3 \\ &\Rightarrow 9 \text{ kN} \\ \frac{b}{3} &\Rightarrow \frac{2}{3} \times 3 \text{ m} = 2 \text{ m} \end{aligned}$$



$$\rightarrow \sum f_x \Rightarrow 0$$

$$\boxed{R_{AH} \Rightarrow 0}$$



$$\Sigma f_y \Rightarrow 0$$

$$R_{AV} + R_B - 9 - 4 \Rightarrow 0$$

$$\boxed{R_{AV} + R_B \Rightarrow 13} \quad \text{--- (1)}$$



$$\Sigma M_A \Rightarrow 0$$

$$\Rightarrow (9 \times 3 + 4 \times 6) - R_B \times 5 \Rightarrow 0$$

$$51 - R_B \times 5 \Rightarrow 0$$

$$R_B \Rightarrow \frac{51}{5} \Rightarrow \boxed{10.2 \text{ kN}}$$

$$R_{AV} + R_B \Rightarrow 13$$

$$R_{AV} + 10.2 \Rightarrow 13$$

$$\boxed{R_{AV} \Rightarrow 2.8 \text{ kN}}$$

Frame $\rightarrow$ A frame may be defined as a structure made up of several bars, riveted or welded together. These bars are made up of angle iron or channel section and are called as members of the frame. A multiforce member is one with three or more forces acting on it, or one with two or more forces and one or more couples acting on it.

Types of frame $\rightarrow$

1) Perfect frame $\rightarrow$ A structural frame that is stable under load imposed upon it from any direction but which would become unstable if one of its members were removed or one of its fixed ends became hinged.

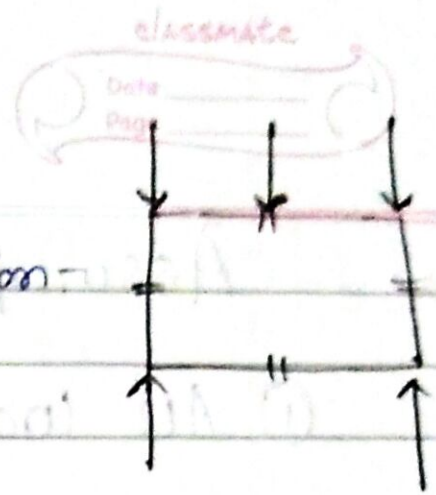
$$n \Rightarrow (2J - 3)$$

$n \Rightarrow$ No. of Members

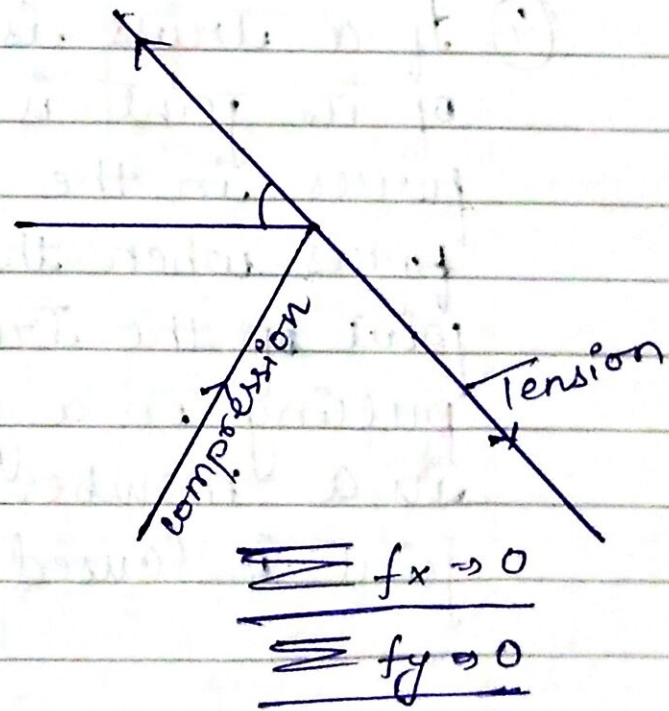
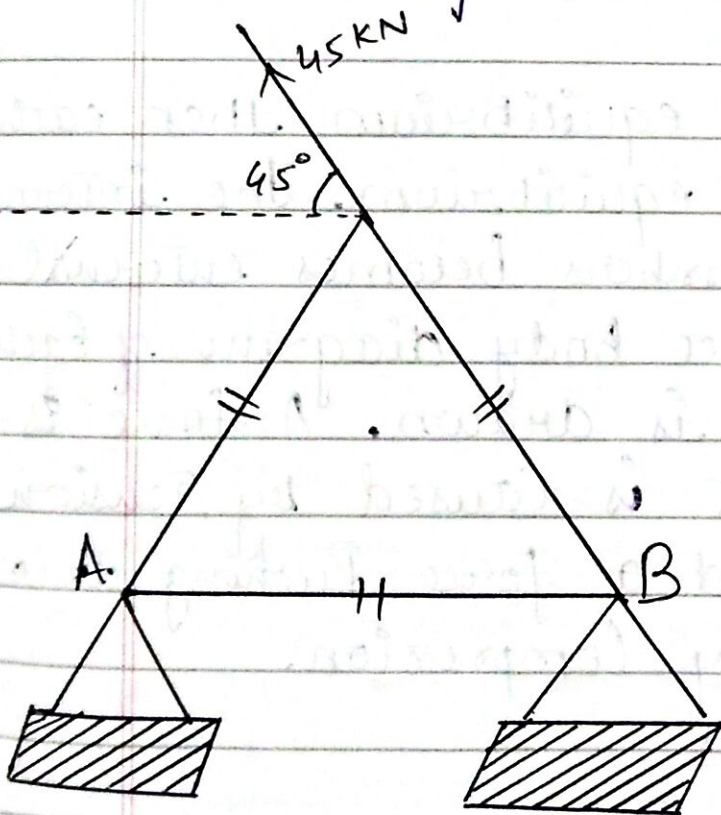
$J \Rightarrow$ No. of Joints

2) Imperfect frame $\rightarrow$ Frame which cannot be analyzed to get internal member force and external support reaction.

It does not satisfy the equation -
 $m \rightarrow 2J - 3$



Truss $\rightarrow$ Truss is a structure that consists of two force members only where the members are organized so that assemblage as a whole behaves as a single object. Truss are widely used in civil engineering application such as bridges, steel building and roof structure.



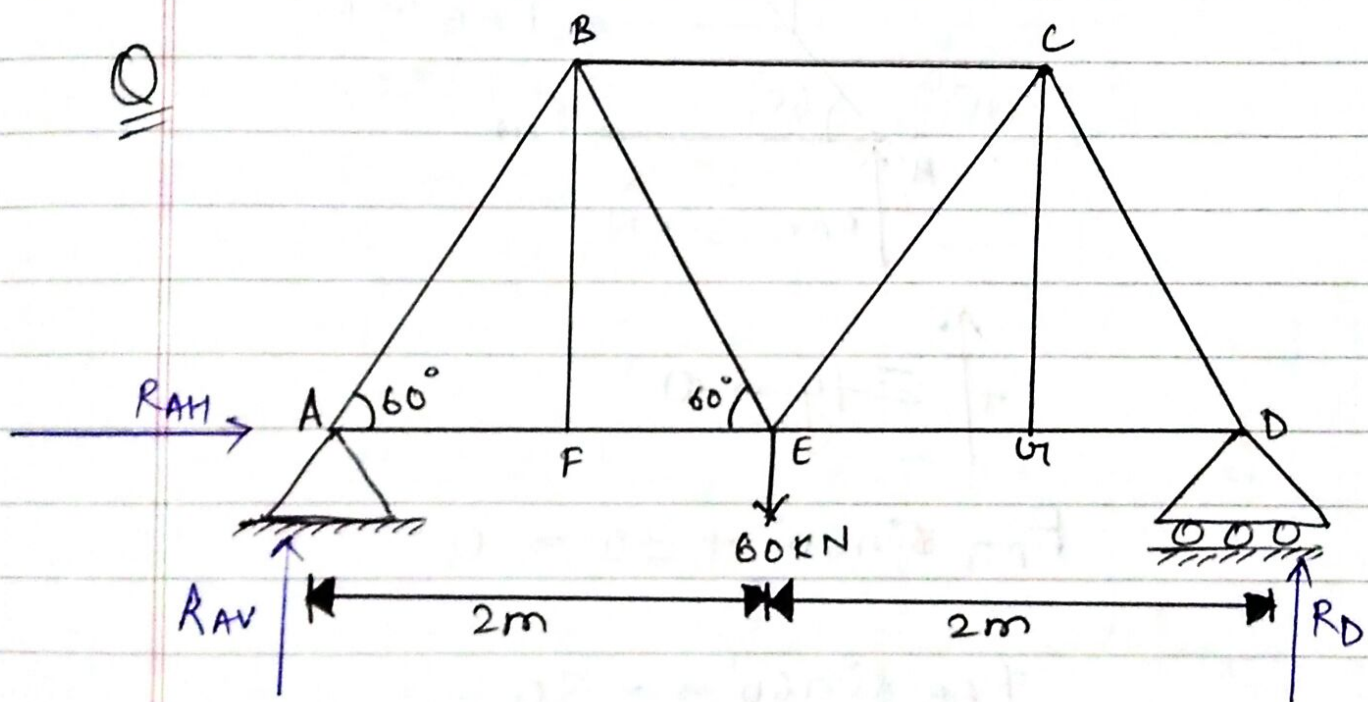
Assumptions for Design :-

- ① All loading are applied at the joint.
- ② The members are joint together by smooth Pins.
- ③ Simple Trusses are composed of Triangular element. The members are assumed to be Pin connected at their ends and load applied at the joint.
- ④ If a Truss is in equilibrium then each of its joint is in equilibrium. The internal forces in the members becomes external forces when the free body diagram of each joint of the Truss is drawn. A force is pulling on a joint is caused by Tension in a member and a force Pushing on a joint is caused by Compression.

Trusses can be solved using two methods :-

- ① Joint Method -
- ② Section Method -

JOINT METHOD PROBLEM.



Sol.

$\rightarrow$

$$\sum F_x = 0$$

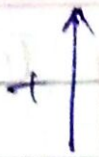
$$R_{AH} = 0$$

$\uparrow$
 $\downarrow$

Moment at Point A

$$60 \times 2 - R_D \times 4 = 0$$

$$R_D = 30 \text{ kN}$$

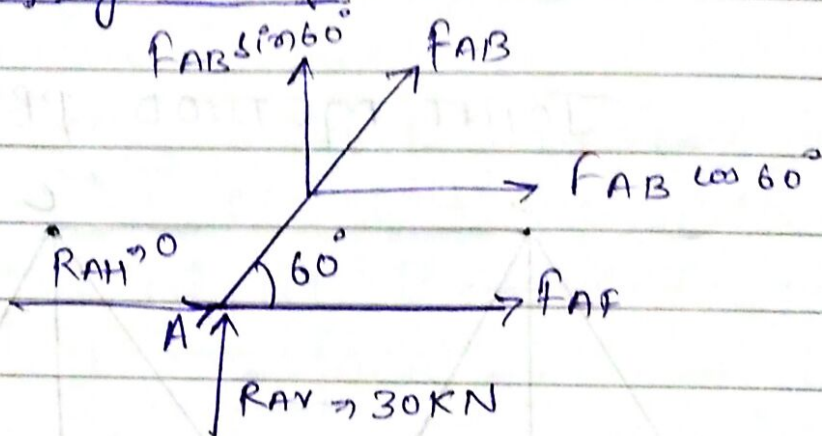


$$\sum f_y = 0$$

$$R_{AV} + 30 - 60 = 0$$

$$R_{AV} = 30 \text{ KN}$$

FBD of joint A

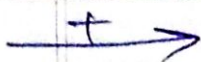


$$\uparrow \sum f_y = 0$$

$$F_{AB} \sin 60^\circ + 30 = 0$$

$$F_{AB} \sin 60^\circ = -30$$

$$F_{AB} = -34.641 \text{ KN} \quad (\text{compression})$$



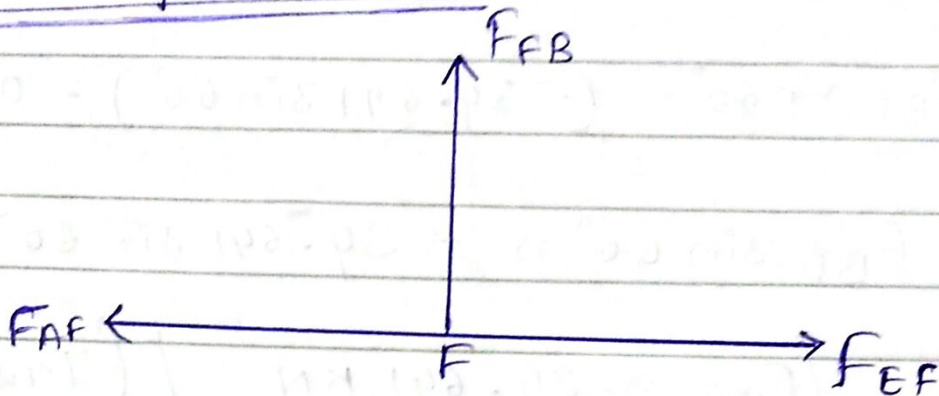
$$\sum f_x = 0$$

$$F_{AF} + F_{AB} \cos 60^\circ = 0$$

$$F_{AF} - 34.641 \cos 60^\circ = 0$$

$$\boxed{F_{AF} \Rightarrow 17.32 \text{ kN}} \text{ (Tension)}$$

FBD of Joint F



$\rightarrow$

$$\sum f_x \Rightarrow 0$$

$$F_{EF} - F_{AF} \Rightarrow 0$$

$$F_{EF} \Rightarrow F_{AF}$$

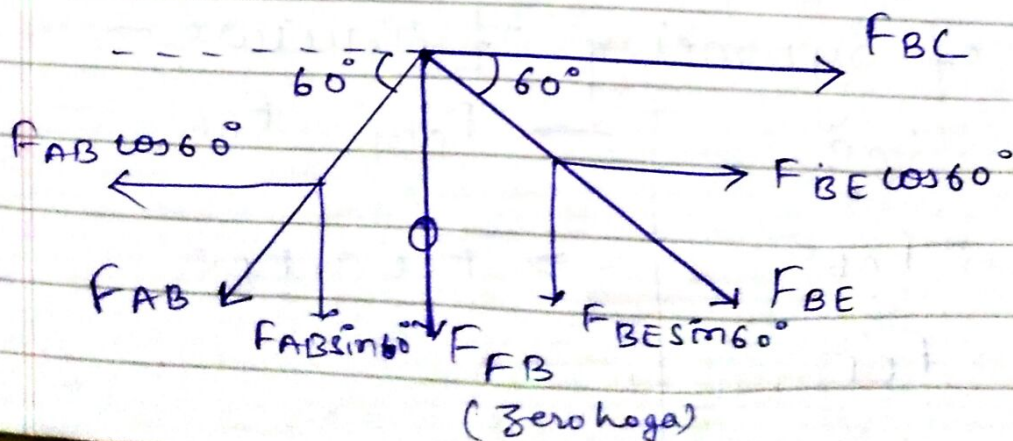
$$\boxed{F_{EF} \Rightarrow 17.32 \text{ kN}} \text{ (T)}$$

$\uparrow$

$$\sum f_y \Rightarrow 0$$

$$\boxed{F_{FB} \Rightarrow 0}$$

F.B.D of joint B



$$+\uparrow \Sigma f_y \Rightarrow 0$$

$$\Rightarrow -F_{BE} \sin 60^\circ - F_{AB} \sin 60^\circ - F_{FB} \Rightarrow 0$$

$$\Rightarrow -F_{BE} \sin 60^\circ - (-34.641 \sin 60^\circ) - 0 \Rightarrow 0$$

$$\Rightarrow F_{BE} \sin 60^\circ \Rightarrow 34.641 \sin 60^\circ$$

$$\boxed{F_{BE} \Rightarrow 34.641 \text{ KN}} \quad (\text{Tension})$$

$$+\rightarrow \Sigma f_x \Rightarrow 0$$

$$\Rightarrow F_{BC} + F_{BE} \cos 60^\circ - F_{AB} \cos 60^\circ \Rightarrow 0$$

$$\Rightarrow F_{BC} + 34.641 \cos 60^\circ - (-34.641) \cos 60^\circ = 0$$

$$F_{BC} + 34.641 \Rightarrow 0$$

$$\boxed{F_{BC} \Rightarrow -34.641 \text{ KN}} \quad (\text{Compression})$$

Due to Symmetry of structure -

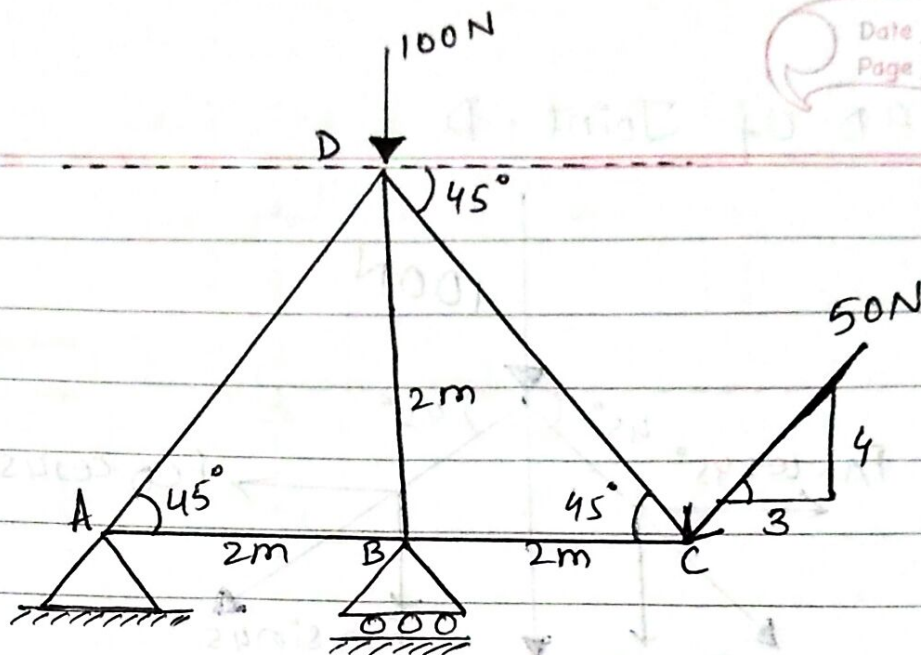
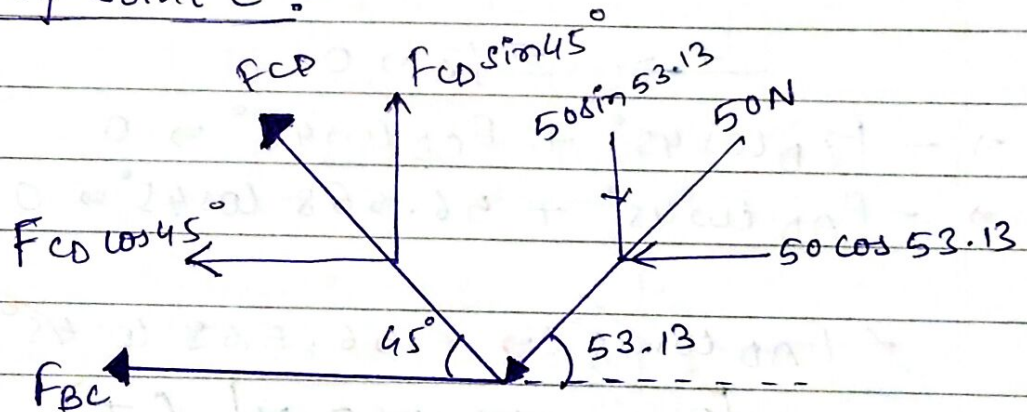
$$\rightarrow F_{CD} \Rightarrow F_{AB}$$

$$\rightarrow F_{CD} \Rightarrow F_{FB}$$

$$\rightarrow F_{GD} \Rightarrow F_{AF}$$

$$\rightarrow F_{CE} \Rightarrow F_{BE}$$

$$\rightarrow F_{HE} \Rightarrow F_{EF}$$

SolⁿF.B.D of Joint C :-

$$+\uparrow \Sigma F_y \Rightarrow 0$$

$$F_{CD} \sin 45^\circ - 50 \sin 53.13 \Rightarrow 0$$

$$F_{CD} \sin 45^\circ - 40 \Rightarrow 0$$

$$\boxed{F_{CD} \Rightarrow 56.568} \text{ (T)}$$

$$+\rightarrow$$

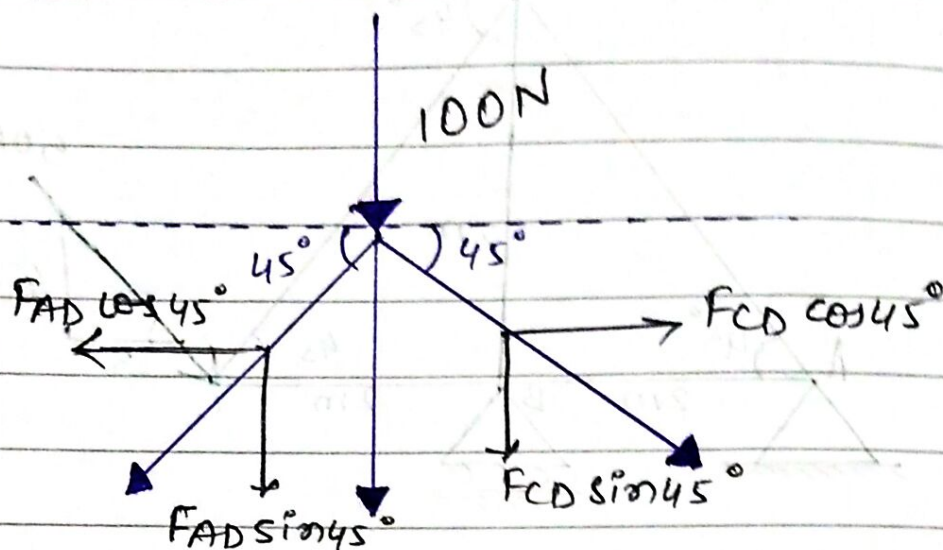
$$\Sigma F_x \Rightarrow 0$$

$$\Rightarrow -F_{BC} - F_{CD} \cos 45^\circ - 50 \cos 53.13 \Rightarrow 0$$

$$\Rightarrow -F_{BC} - 56.568 \cos 45^\circ - 50 \cos 53.13 \Rightarrow 0$$

$$-F_{BC} - 70 \Rightarrow 0$$

$$\boxed{F_{BC} \Rightarrow -70 \text{ N}} \text{ (C)}$$

FBD of Joint D

$$\rightarrow \Sigma f_x \Rightarrow 0$$

$$\Rightarrow -F_{AD} \cos 45^\circ + F_{CD} \cos 45^\circ \Rightarrow 0$$

$$\Rightarrow -F_{AD} \cos 45^\circ + 56.568 \cos 45^\circ \Rightarrow 0$$

$$\cancel{F_{AD} \cos 45^\circ} \Rightarrow \cancel{56.568 \cos 45^\circ}$$

$$\boxed{F_{AD} \Rightarrow 56.568 \text{ N}} \quad (T)$$

$$\uparrow \Sigma f_y \Rightarrow 0$$

$$\Rightarrow -F_{BD} - F_{AD} \sin 45^\circ - F_{CD} \sin 45^\circ - 100 \Rightarrow 0$$

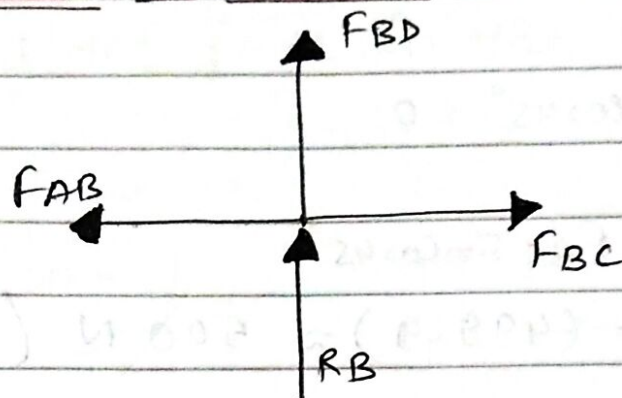
$$\Rightarrow -F_{BD} - 56.568 \sin 45^\circ - 56.568 \sin 45^\circ - 100 \Rightarrow 0$$

$$-F_{BD} - 180 \Rightarrow 0$$

$$-F_{BD} \Rightarrow 180$$

$$\boxed{F_{BD} \Rightarrow -180 \text{ N}} \quad (C)$$

F.B.D. of Joint B



$$\rightarrow \sum f_x \Rightarrow 0$$

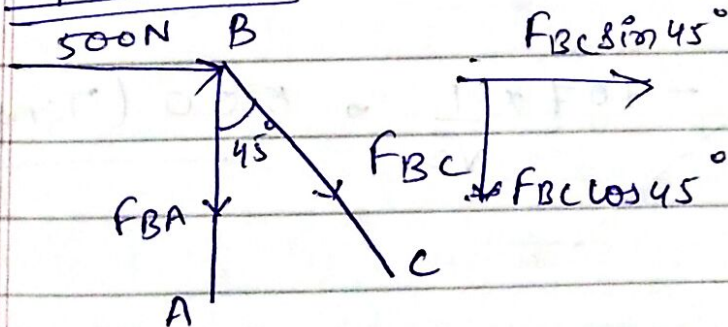
$$F_{BC} - F_{AB} \Rightarrow 0$$

$$F_{BC} \Rightarrow F_{AB}$$

$$\boxed{F_{AB} \Rightarrow -70 \text{ N}} \quad (C)$$

Q = find the force in the members — 500 N

Sol At point B :-

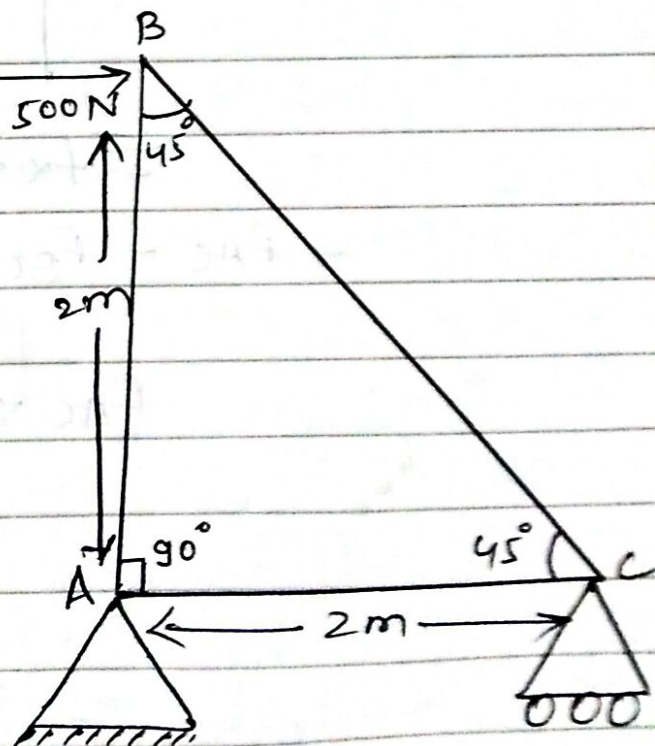


$$\sum f_x \Rightarrow 0$$

$$500 + F_{BC} \sin 45^\circ \Rightarrow 0$$

$$F_{BC} \Rightarrow -500\sqrt{2}$$

$$\Rightarrow -707.106 \text{ N (compression)}$$



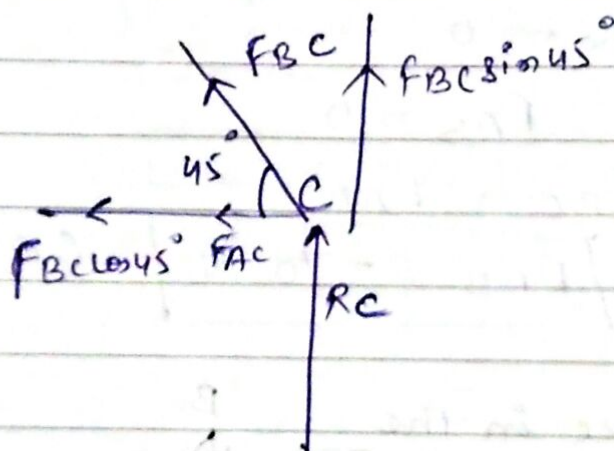
$$\sum f_y \Rightarrow 0$$

$$-F_{BA} - F_{BC} \cos 45^\circ \Rightarrow 0$$

$$F_{BA} \Rightarrow -F_{BC} \cos 45^\circ$$

$$\Rightarrow -(499.9) \approx 500 \text{ N (Tension)}$$

AT point C



$$\sum f_x \Rightarrow 0$$

$$-F_{AC} - F_{BC} \cos 45^\circ = 0$$

$$F_{AC} \Rightarrow -707 \times \frac{1}{\sqrt{2}} \Rightarrow 500 \text{ (Tension)}$$

Q Find the force in the Member —

Sol. $\rightarrow \sum f_x \Rightarrow 0$
 $\boxed{R_{AH} \Rightarrow 0}$

$\uparrow \sum f_y \Rightarrow 0$
 $R_{AV} + R_B - 10 \Rightarrow 0$
 $R_{AV} + R_B \Rightarrow 10 \text{ — (1)}$

$\curvearrowright \sum M_A \Rightarrow 0$
 $10 \times 3 - R_B \times 6 \Rightarrow 0$
 $30 - R_B \times 6 \Rightarrow 0$

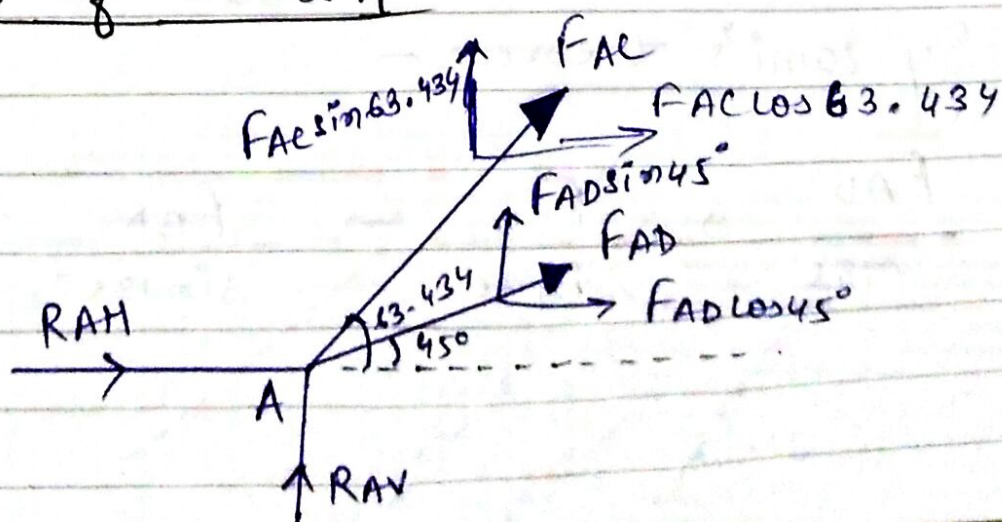
$\boxed{R_B \Rightarrow 5 \text{ KN}}$

Put the value of R_B in eq — (1)

$\boxed{R_{AV} \Rightarrow 5 \text{ KN}}$

In $\triangle AED$
 $\tan \theta \Rightarrow \frac{6}{3}$
 $\tan \theta = 2$
 $\theta = \tan^{-1}(2)$
 $\theta = 63.434^\circ$

F.B.D of Joint A



→

$$\sum f_x \Rightarrow 0$$

$$F_{AD} \cos 45^\circ + F_{AE} \cos 63.434 \Rightarrow 0 \quad \text{--- (2)}$$

↑

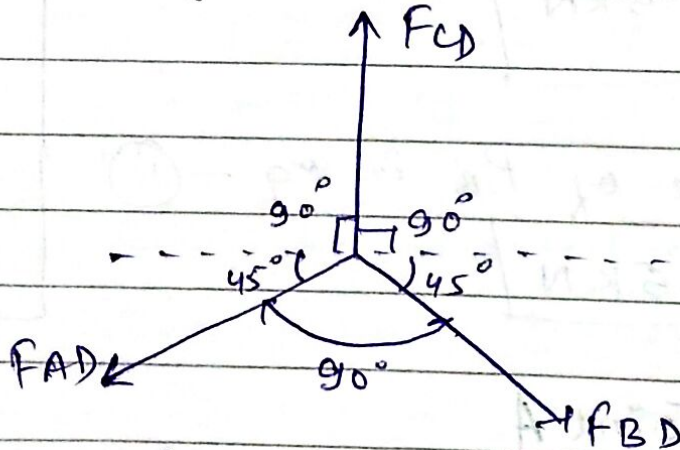
$$\sum f_y \Rightarrow 0$$

$$F_{AD} \sin 45^\circ + F_{AE} \sin 63.434 + 5 \Rightarrow 0 \quad \text{--- (3)}$$

Solve (2) & (3)

$$F_{AD} \Rightarrow 7.071 \text{ KN (T)}$$

$$F_{AE} \Rightarrow -11.18 \text{ KN (C)}$$

FBD of Joint D

By Lami's theorem -

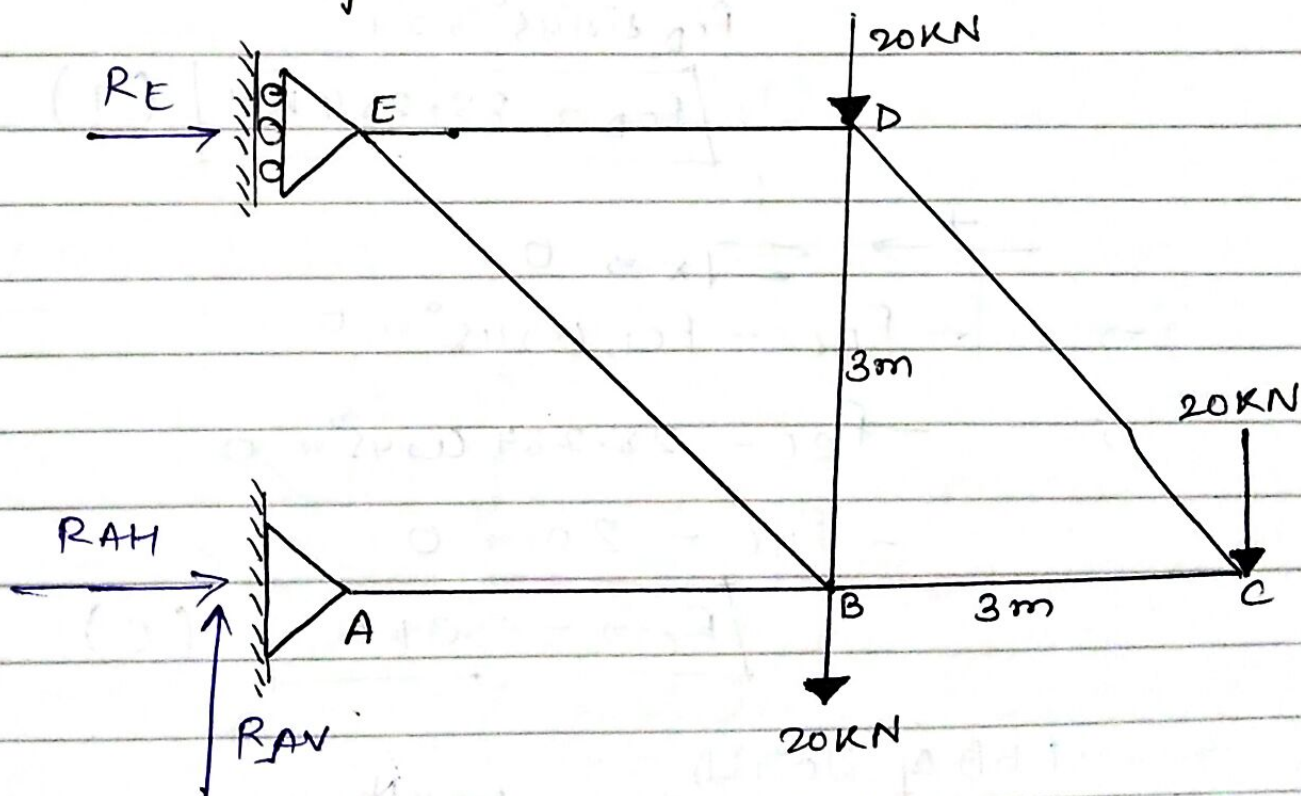
$$\frac{F_{AD}}{\sin 135^\circ} \Rightarrow \frac{F_{CD}}{\sin 90^\circ} \Rightarrow \frac{F_{BD}}{\sin 135^\circ}$$

$$\therefore \frac{F_{AD}}{\sin 135^\circ} = \frac{F_{CD}}{\sin 90^\circ}$$

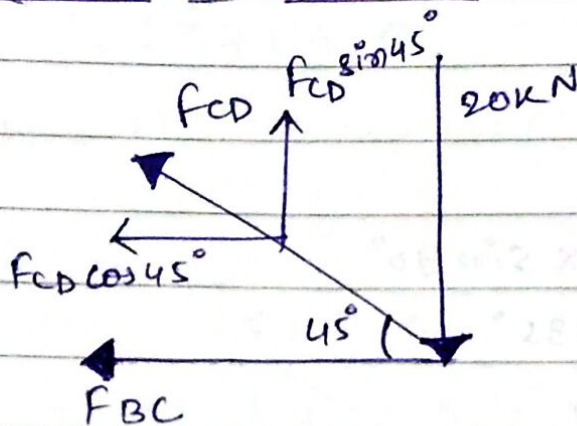
$$F_{CD} = \frac{7.071 \times \sin 90^\circ}{\sin 135^\circ}$$

$$F_{CD} = 10 \text{ kN (T)}$$

Q Find the force in all the members -



① → ② → ③ → ④ → ⑤

FBD of Joint C

$$\uparrow \sum f_y \Rightarrow 0$$

$$F_{CD} \sin 45^\circ - 20 \Rightarrow 0$$

$$F_{CD} \sin 45^\circ \Rightarrow 20$$

$$\boxed{F_{CD} \Rightarrow 28.284 \text{ kN}} \quad (T)$$

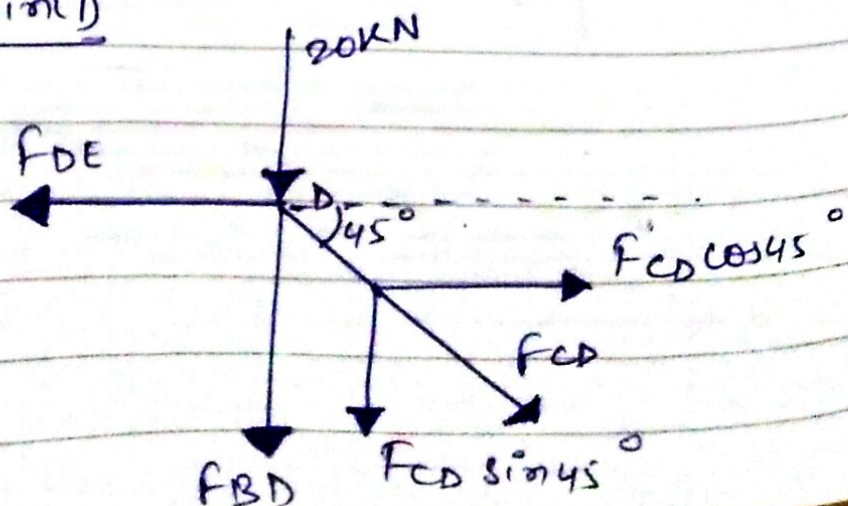
$$\rightarrow \sum f_x \Rightarrow 0$$

$$\Rightarrow -F_{BC} - F_{CD} \cos 45^\circ \Rightarrow 0$$

$$\Rightarrow -F_{BC} - 28.284 \cos 45^\circ \Rightarrow 0$$

$$-F_{BC} - 20 \Rightarrow 0$$

$$\boxed{F_C \Rightarrow -20 \text{ kN}} \quad (C)$$

FBD of Joint D

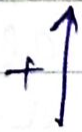


$$\sum F_x \Rightarrow 0$$

$$-F_{DE} + 28.284 \cos 45^\circ \Rightarrow 0$$

$$-F_{DE} + 20 \Rightarrow 0$$

$$\boxed{F_{DE} \Rightarrow 20 \text{ kN}} \quad (T)$$



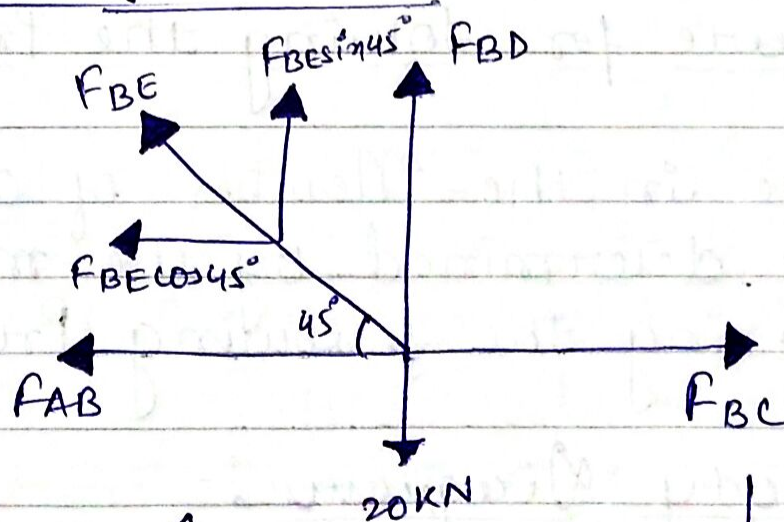
$$\sum F_y \Rightarrow 0$$

$$-F_{BD} - 28.284 \sin 45^\circ - 20 \Rightarrow 0$$

$$-F_{BD} - 40 \Rightarrow 0$$

$$\boxed{F_{BD} \Rightarrow -40 \text{ kN}} \quad (C)$$

FBD of Joint (B)



$$+ \uparrow \sum F_y \Rightarrow 0$$

$$F_{BE} \sin 45^\circ + F_{BD} - 20 \Rightarrow 0$$

$$F_{BE} \sin 45^\circ + 40 - 20 \Rightarrow 0$$

$$F_{BE} \sin 45^\circ \Rightarrow 60$$

$$\boxed{F_{BE} \Rightarrow 84.852 \text{ kN}} \quad (T)$$

$$+ \rightarrow \sum F_x \Rightarrow 0$$

$$-F_{AB} + F_{BC} - F_{BE} \cos 45^\circ \Rightarrow 0$$

$$-F_{AB} - 20 - 84.852 \cos 45^\circ \Rightarrow 0$$

$$-F_{AB} - 80 \Rightarrow 0$$

$$\boxed{F_{AB} \Rightarrow -80 \text{ kN}} \quad (C)$$

2) METHOD OF SECTION :-

Important Points :-

- If a Truss is in equilibrium then Each of its segment is in Equilibrium. The internal forces in the Members becomes external forces with the free body diagram of a segment of the Truss is drawn. If force Pulling on a member Causes Tension in the Member and a force Pushing on a Member Causes compression.
- Procedure for solving the Problem :-

The force in the Member of a Truss may be determined by the method of section using the following Procedure:-

Free Body Diagram :-

- ① Make a decision on how To cut or section the Truss through the Members where the force are to be determined.

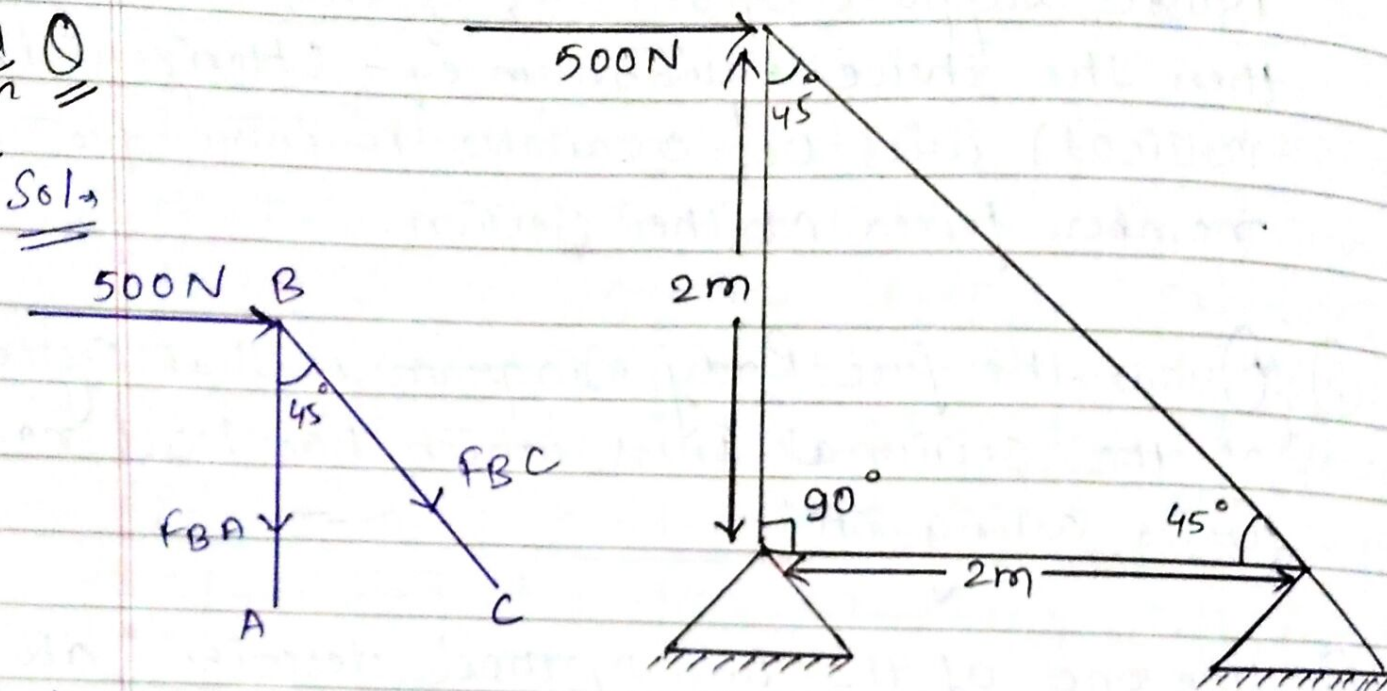
- ② Before isolating the appropriate section, it may be first necessary to determine the Trusses support reactions. If this is done, then the three equilibrium eq - (Horizontal, vertical) will be available to solve for member forces at the section.
- ③ Draw the free Body Diagram of that Segment of the sectional Truss which has least no. of forces acting on it.
- ④ Use one of the two Method described above for establishing sense of the Unknown member forces.

Equation of Equilibrium

- ① Moment should be Sum about a Point that lies at a intersection of the line of action of two unknown forces so that the third unknown forces can be determined directly from the Moment Eq.
- ② If two of the Unknown forces are Parallel, forces may be Sum Perpendicular to the

direction of this Unknown to determine directly the third Unknown forces.

Mid
sem
qs
Soln



$$\rightarrow \sum f_x \Rightarrow 0$$

$$500 + F_{BC} \sin 45^\circ \Rightarrow 0$$

$$F_{BC} \Rightarrow -500\sqrt{2}$$

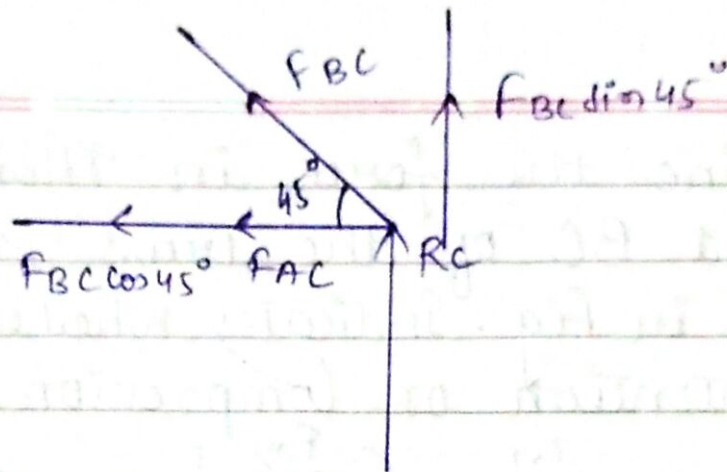
$$F_{BC} \Rightarrow -707.106 \text{ N (Compression)}$$

$$\uparrow \sum f_y \Rightarrow 0$$

$$-F_{BA} - F_{BC} \cos 45^\circ \Rightarrow 0$$

$$F_{BA} \Rightarrow -F_{BC} \cos 45^\circ$$

$$F_{BA} \Rightarrow -499.9 \approx 500 \text{ N (Tension)}$$

AT Point C

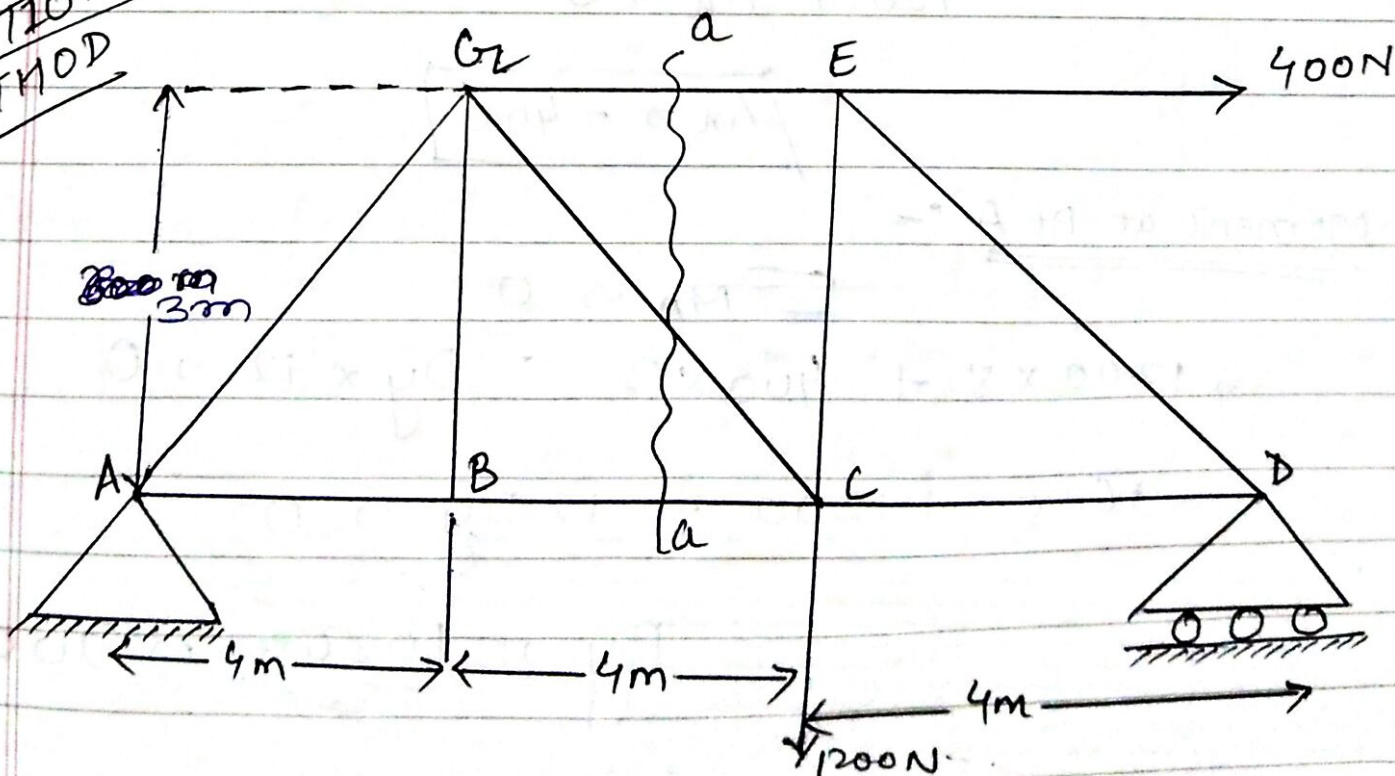
$$\sum f_x \Rightarrow 0$$

$$\Rightarrow -F_{AC} - F_{BC} \cos 45^\circ \Rightarrow 0$$

$$-F_{AC} \Rightarrow F_{BC} \cos 45^\circ$$

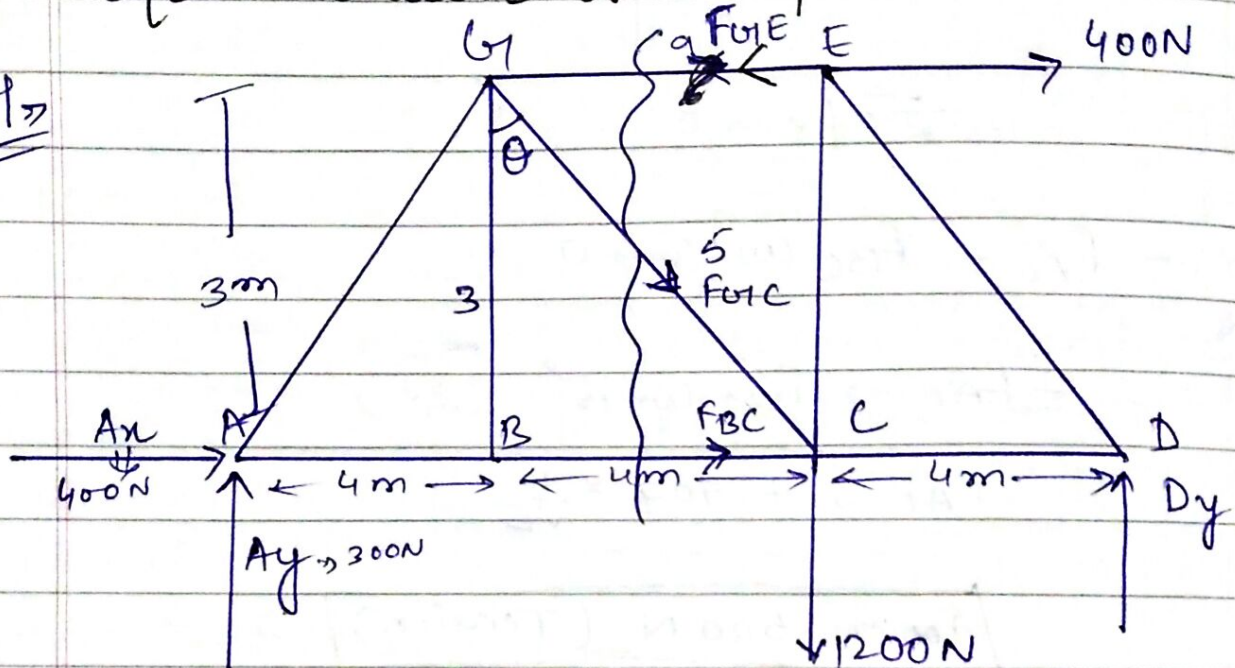
$$F_{AC} \Rightarrow -707 \times \frac{1}{\sqrt{2}}$$

$$F_{AC} \Rightarrow 500 \text{ N (Tension)}$$

SECTION METHOD

Determine the force in Member GE & GC and BC of the Truss which is shown in fig. Indicates whether the Members are tension or compression.

Solⁿ



$$\sum f_x = 0$$

$$400 + A_x = 0$$

$$A_x = -400$$

Moment at Pt A :-

$$\sum M_A = 0$$

$$\Rightarrow 1200 \times 8 + 400 \times 3 - D_y \times 12 = 0$$

$$\Rightarrow 9600 + 1200 - 12D_y = 0$$

$$D_y = \frac{10800}{12} = 900 \text{ N}$$

+↑

$$\sum f_y = 0$$

$$D_y + A_y - 1200 = 0$$

$$900 + A_y - 1200 = 0$$

$$\boxed{A_y = 300 \text{ N}}$$

Moment at Point G

$$\Rightarrow (300 \times 4) + (400 \times 3) - F_{BC} \times 3 = 0$$

$$F_{BC} = \frac{2400}{3} = 800 \text{ (T)}$$

Moment at Point C

$$(300 \times 8) - F_{GE} \times 3 = 0$$

$$F_{GE} = \frac{2400}{3} = 800 \text{ (T)}$$

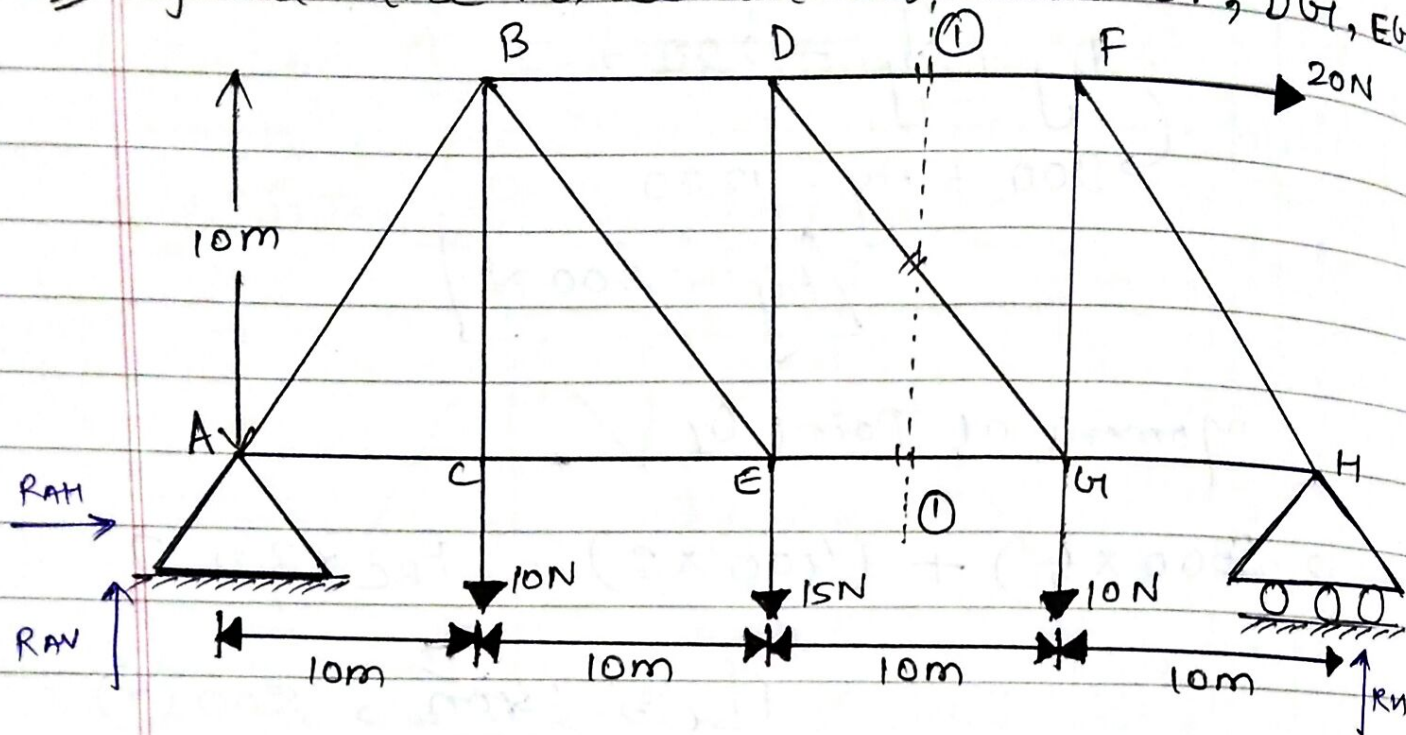
Vertical forces

$$300 - F_{GC} \cos \theta = 0 \quad \rightarrow \frac{3}{5}$$

$$300 - F_{GC} \times \frac{3}{5} = 0$$

$$\boxed{F_{GC} = 500 \text{ N}}$$

Q Find the force in Members DF, DG, EG.



Sol's

$$\rightarrow \sum f_x \Rightarrow 0$$

$$R_{AH} + 20 \Rightarrow 0$$

$$\boxed{R_{AH} \Rightarrow -20 \text{ N}}$$

$$\uparrow \sum f_y \Rightarrow 0$$

$$R_{AV} + R_H \Rightarrow 10 + 15 + 10$$

$$\boxed{R_{AV} + R_H \Rightarrow 35}$$

$$\curvearrowright \sum M_A \Rightarrow 0$$

$$\Rightarrow (10 \times 10 + 15 \times 20 + 10 \times 30 + 20 \times 10) - R_H \times 40 \Rightarrow 0$$

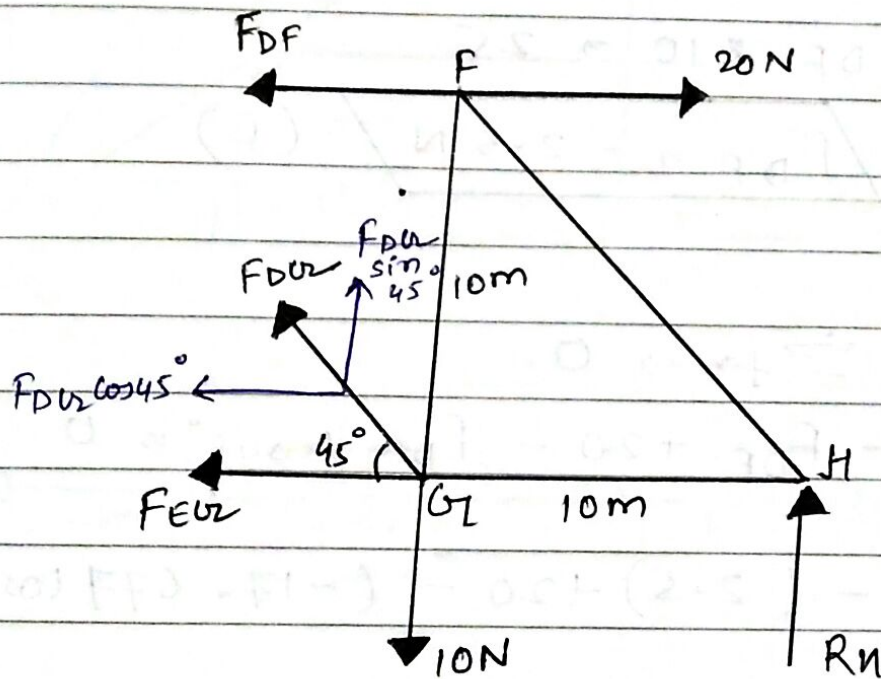
$$\Rightarrow 900 - 40 \times R_H \Rightarrow 0$$

$$\boxed{R_H \Rightarrow 22.5}$$

$$R_{AV} + 22.5 \Rightarrow 35$$

$$R_{AV} \Rightarrow 12.5 \text{ N}$$

NOTE-2 {When you cut the Trusses using Sectional Plane then it should be cut three or must Unknown Members}.



$$+\uparrow \Sigma f_y \Rightarrow 0$$

$$F_{DG} \sin 45^\circ - 10 + R_H \Rightarrow 0$$

$$F_{DG} \sin 45^\circ - 10 + 22.5 \Rightarrow 0$$

$$F_{DG} \sin 45^\circ + 12.5 \Rightarrow 0$$

$$F_{DG} \sin 45^\circ \Rightarrow -12.5$$

$$F_{DG} \Rightarrow -17.677 \text{ N (C)}$$

↑
↓

$$\sum M_G \Rightarrow 0$$

$$- F_{DF} \times 10 + 20 \times 10 + R_H \times 10 \Rightarrow 0$$

$$- F_{DF} \times 10 + 200 - 22.5 \times 10 \Rightarrow 0$$

$$- F_{DF} \times 10 \Rightarrow 25$$

$$\boxed{F_{DF} \Rightarrow -2.5 \text{ N}} \quad (C)$$

→

$$\sum F_x \Rightarrow 0$$

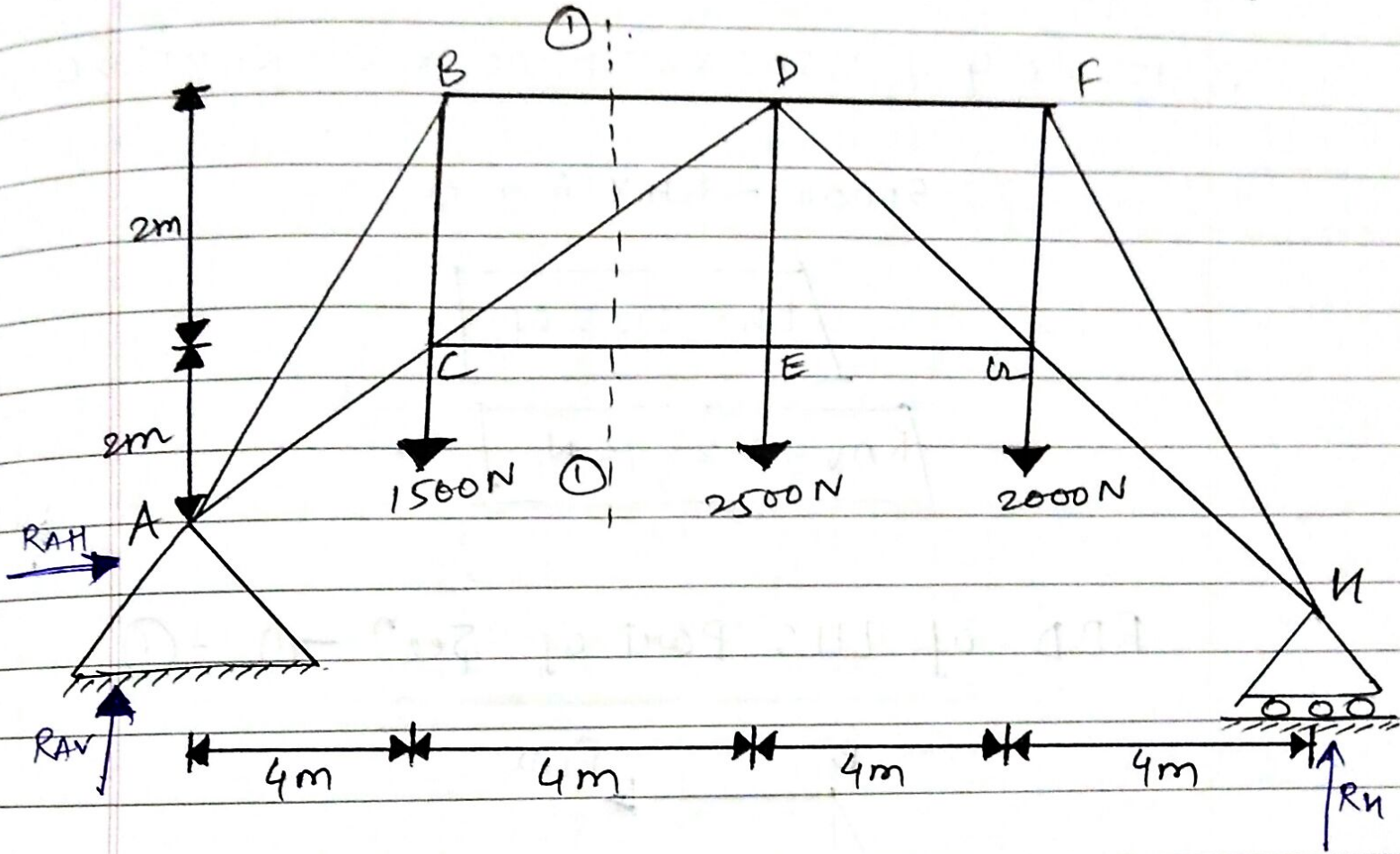
$$\Rightarrow -F_{EG} - F_{DF} + 20 - F_{DG} \cos 45^\circ \Rightarrow 0$$

$$\Rightarrow -F_{EG} - (-2.5) + 20 - (-17.677 \cos 45^\circ) \Rightarrow 0$$

$$\Rightarrow -F_{EG} + 35 \Rightarrow 0$$

$$\boxed{F_{EG} \Rightarrow 35 \text{ N}} \quad (T)$$

Q Find the force in Members BD, CD & CE.



Sol.

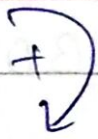
$$t \rightarrow \sum f_x \rightarrow 0$$

$$RAH \Rightarrow 0$$

$$+ \uparrow \sum f_y = 0$$

$$R_A + R_H - 1500 - 2500 - 2000 = 0$$

$$R_{AV} + R_H - 6000 = 0 \quad \text{--- (1)}$$



$$\sum M_A \Rightarrow 0$$

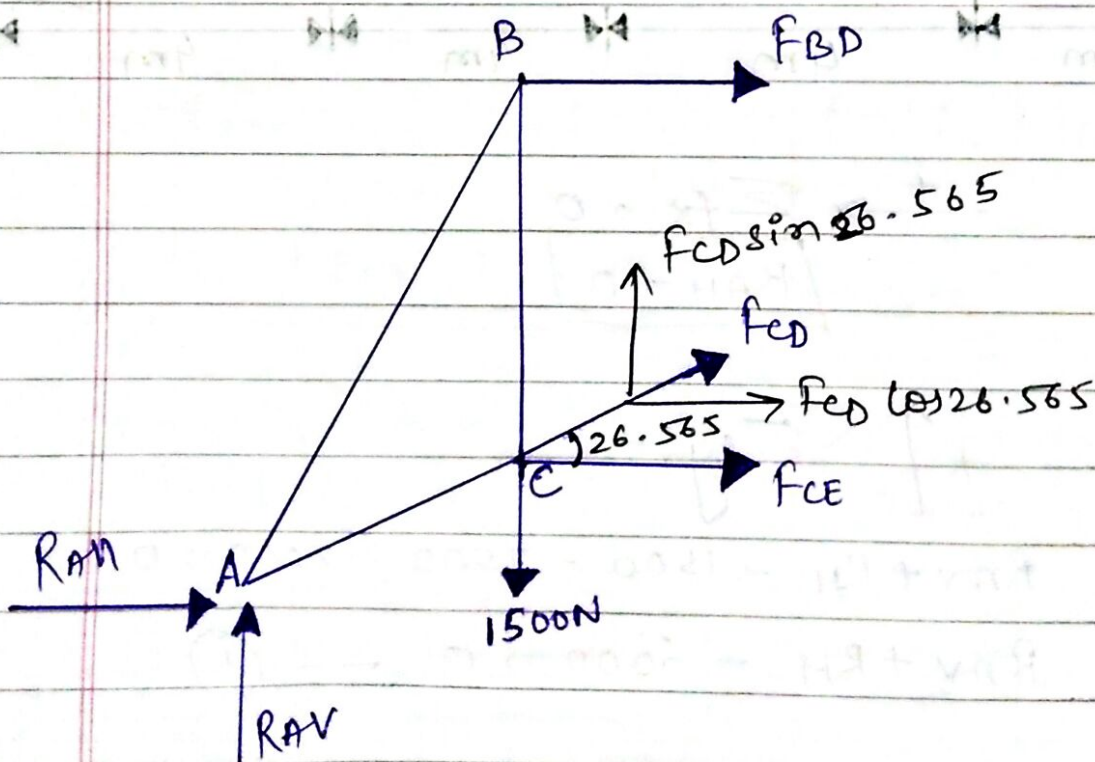
$$\Rightarrow 1500 \times 4 + 2500 \times 8 + 2000 \times 12 - R_H \times 16 \Rightarrow 0$$

$$50000 - R_H \times 16 \Rightarrow 0$$

$$R_H \Rightarrow 3125 \text{ N}$$

$$R_{AV} \Rightarrow 2875 \text{ N}$$

FBD of LHS part of secⁿ - ① - ①



$$\tan \theta = \frac{2}{4}$$

$$\theta = \tan^{-1}(0.5)$$

$$\theta = 26.565^\circ$$

$$+\uparrow \Sigma f_y \Rightarrow 0$$

$$\Rightarrow F_{CD} \sin 26.565 - 1500 + 2875 \Rightarrow 0$$

$$\Rightarrow F_{CD} \sin 26.565 + 1375 \Rightarrow 0$$

$$F_{CD} \sin 26.565 \Rightarrow -1375$$

$$\boxed{F_{CD} \Rightarrow 3074.598 \text{ N}} \quad (C)$$

$$+\downarrow \Sigma M_C \Rightarrow 0$$

$$F_{BD} \times 2 + 2875 \times 4 \Rightarrow 0$$

$$\boxed{F_{BD} \Rightarrow -5750 \text{ N}} \quad (C)$$

$$+\rightarrow \Sigma f_x \Rightarrow 0$$

$$\Rightarrow F_{CE} + F_{BD} + F_{CD} \cos 26.565 \Rightarrow 0$$

$$\Rightarrow F_{CE} - 5750 - 3074.598 \cos 26.565 \Rightarrow 0$$

$$F_{CE} - 8500.005 \Rightarrow 0$$

$$\boxed{F_{CE} \Rightarrow 8500.005 \text{ N}} \quad (T)$$

20.01.24

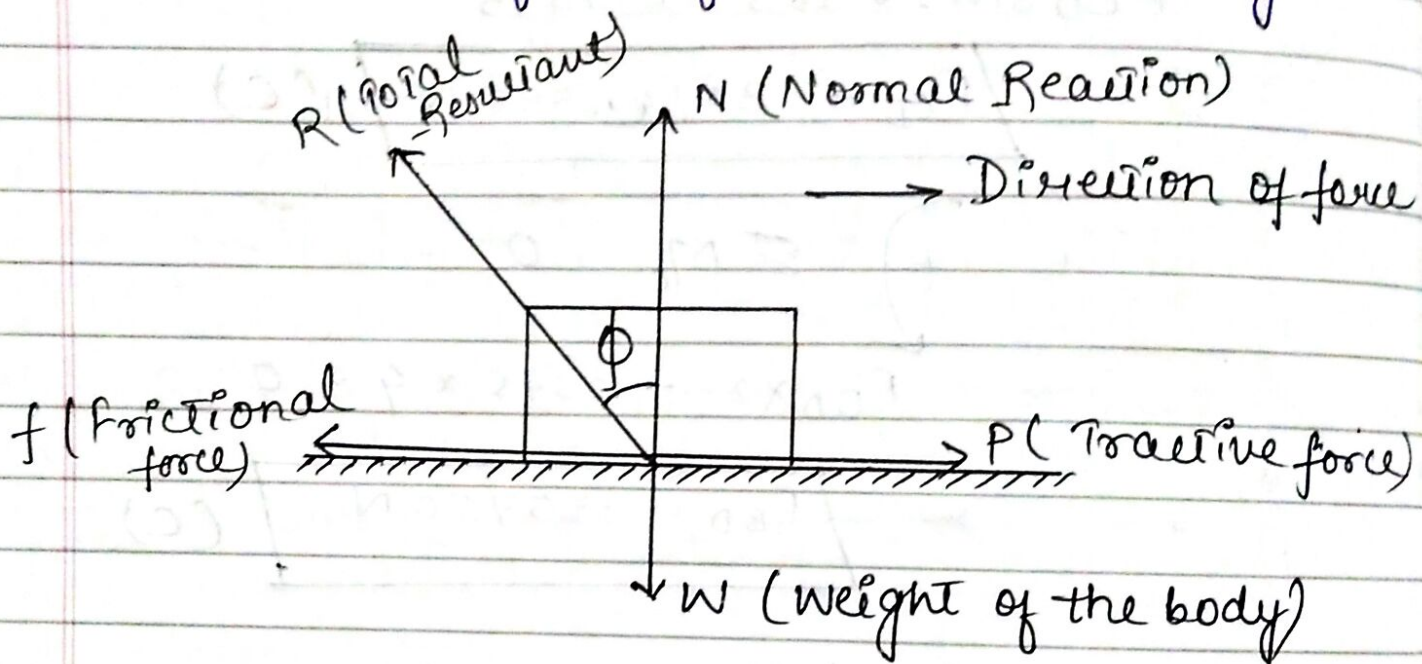
FRICTION

classmate

Date

Page

Force of friction or frictional force may be defined as the opposing force which is caused into a plane in between surface of contact of two bodies, when one body moves over the surface of the another body.



Characteristics of frictional force -

- The force of friction or frictional force entails the -
 - a) It is self-adjusting
 - b) It always act in a direction opposite to the motion.
 - c) It is passive force (false force)

Types of Friction

Friction may be classified in two ways:-

- 1) Friction in Unlubricated surface -
- 2) Friction in lubricated surface

1) Friction in Unlubricated surface → The friction that exist between two lubricated surfaces is called as solid friction or dry friction -

It may have two types -

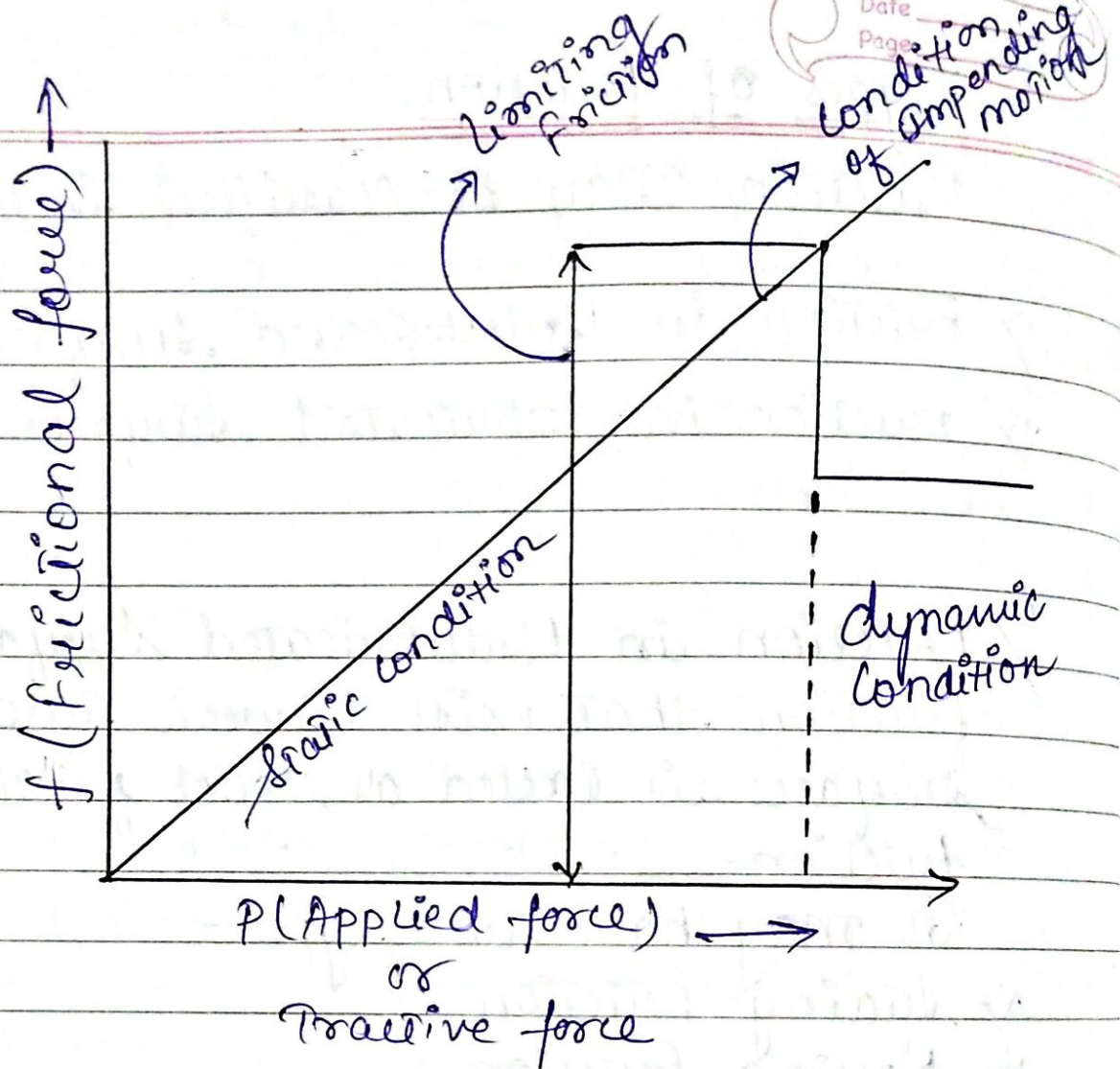
- a) Sliding Friction
- b) Rolling Friction

Sliding Friction → The friction that exists when one surface slides over the other is called as sliding friction.

Rolling Friction → The friction that exists between two surfaces separated by balls or rollers is called the Rolling friction.

NOTE -

It always remember that rolling friction is always less than sliding friction.



Static friction → The static friction is the friction offered by the surface subjected to external forces until there is a no motion between them.

Dynamic or Kinetic friction → The dynamic friction is the friction experienced by the body when it is in motion. It is also known as Kinetic friction and it is always less than static friction.

Limiting friction $\rightarrow$ Limiting force or friction may be defined as the Maximum value of friction force which exist when a body just begin to slide over the surface of other body.

Laws of friction

- 1) Laws of static friction $\rightarrow$ The frictional force always act in a direction opposite to that in which the body tends to move.
- a) The frictional force is directly proportional to ^{the} Normal reaction between the surfaces.
- c) The frictional force depend upon the Nature of surface in contact.
- d) The frictional force is independent of area & shape of the contacting surfaces.
- 2) Laws of Dynamic & Kinetic friction $\rightarrow$
 - a) The frictional force always act in a direction opposite to that in which the body moves.

b) The frictional force is directly Proportional to the Normal reaction between two contacting surfaces.

c) The magnitude of force of Kinetic friction bears a constant ratio to the Normal reaction.

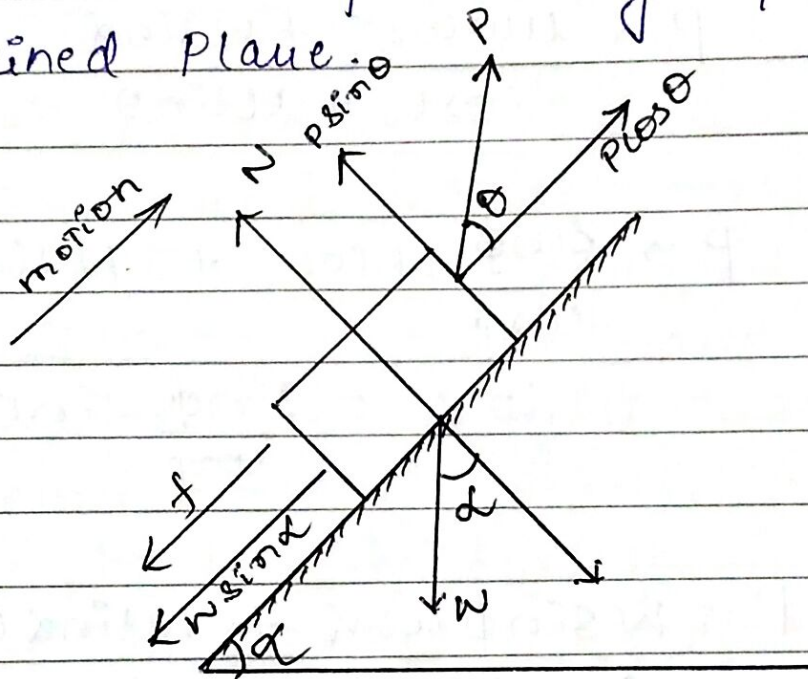
d) The frictional force remains constant for moderate speed but it decreases slightly with the increase of speed.

Angle of friction $\rightarrow$ It is defined as the angle which is resultant of normal reaction (N) and limiting force of friction (f) makes with the normal (N) which is denoted by -

$$\tan \phi \Rightarrow \mu = \frac{f}{N} \quad \mu \Rightarrow \tan \phi$$

Coefficient of friction (μ) $\rightarrow$ It is defined as the ratio of limiting force of friction to the normal reaction between two bodies.

Angle of Repose (α) $\rightarrow$ The angle α is called as angle of Repose (α) and it is equal to the angle of friction with the body is in the condition of limiting equilibrium on an inclined plane.



$$P \sin \theta + N - W \cos \alpha = 0 \quad \text{--- (i)}$$

$$N = W \cos \alpha - P \sin \theta$$

$$F + W \sin \alpha - P \cos \theta = 0$$

$$F + W \sin \alpha = P \cos \theta$$

$$\mu N = P \cos \theta - W \sin \alpha$$

Put the value of N from eq (i)

$$\Rightarrow \mu (W \cos \alpha - P \sin \theta) = P \cos \theta - W \sin \alpha$$

$$\Rightarrow \mu W \cos \alpha - \mu P \sin \theta = P \cos \theta - W \sin \alpha$$

$$\Rightarrow \mu W \cos \alpha + W \sin \alpha \Rightarrow P \cos \theta + \mu P \sin \theta$$

$$\mu W \cos \alpha + W \sin \alpha \Rightarrow P (\cos \theta + \mu \sin \theta)$$

$$P \Rightarrow \frac{\mu W \cos \alpha + W \sin \alpha}{\cos \theta + \mu \sin \theta}$$

$$\mu \Rightarrow \tan \phi$$

$$\mu \Rightarrow \frac{\sin \phi}{\cos \phi}$$

$$P \Rightarrow \frac{\sin \phi \cdot W \cos \alpha + W \sin \alpha}{\cos \theta + \frac{\sin \phi}{\cos \phi} \sin \theta}$$

$$P \Rightarrow \frac{W \sin \phi \cos \alpha + W \sin \alpha \cos \phi}{\cos \theta \cos \phi + \sin \phi \sin \theta}$$

$$P \Rightarrow \frac{W (\sin \phi \cos \alpha + \sin \alpha \cos \phi)}{\cos (\theta - \phi)}$$

$$P \Rightarrow \frac{W \sin (\alpha + \phi)}{\cos (\theta - \phi)}$$

Case - II

$$\theta \Rightarrow 0^\circ$$

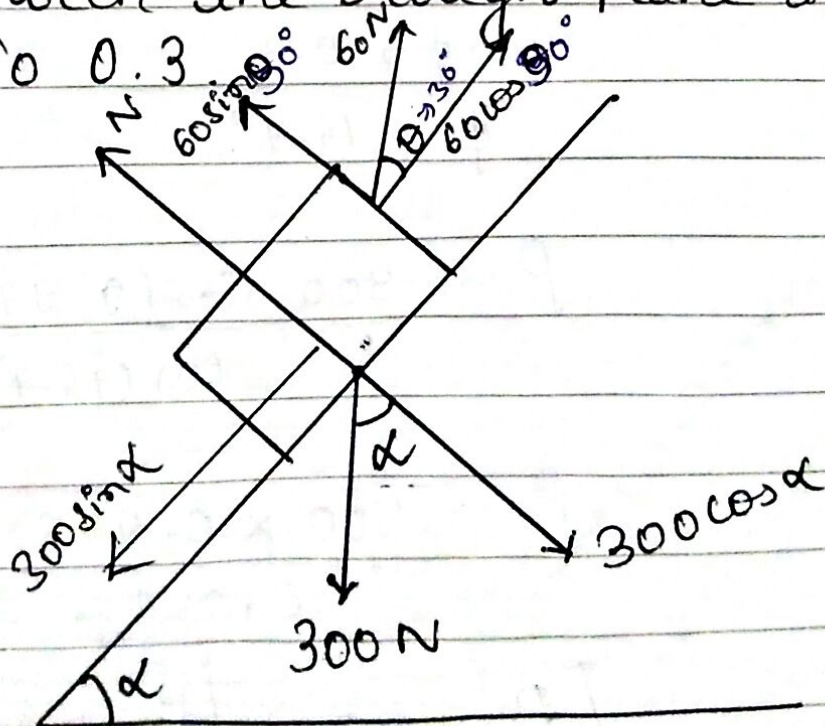
$$P \Rightarrow \frac{W \sin (\alpha + \phi)}{\cos \phi}$$

$$[\cos (-\theta) \Rightarrow \cos \theta]$$

Case III :- $\mu = 0$, $\phi = 0$

$$P = \frac{W \sin \alpha}{\cos \theta}$$

Q Find the force required to move a load of 300 N up a rough inclined Plane, the force being applied parallel to the plane. The inclination of the Plane is such that when the same body is kept on a ~~Perf~~ Perfectly smooth Plane inclined at that angle, a force of 60 N applied at an inclination of 30° to the Plane keeps the same in equilibrium. Assume that co-efficient of friction between the rough Plane and load is equal to 0.3



Sol.

$$W \sin \alpha \Rightarrow 60 \cos 30^\circ$$

$$300 \sin \alpha \Rightarrow 60 \cos 30^\circ$$

$$\sin \alpha \Rightarrow \frac{60 \cos 30^\circ}{300} \Rightarrow 0.1732$$

$$\alpha \Rightarrow \sin^{-1}(0.1732)$$

$$\alpha \Rightarrow 9.97^\circ$$

When force is applied Parallel To Rough Plane.
 & Resolving the forces Parallel & Perpendicular -

Case 1: $\theta = 0^\circ$

$$P \Rightarrow \frac{W \sin(\alpha + \phi)}{\cos \phi}$$

Given - $W = 300 \text{ N}$

$$\mu \Rightarrow 0.3$$

$$\tan \phi \Rightarrow 0.3$$

$$\phi \Rightarrow 16.7^\circ$$

$$P \Rightarrow \frac{300 \sin(9.97^\circ + 16.7^\circ)}{\cos(16.7^\circ)}$$

$$P \Rightarrow \frac{300 \times 0.448}{0.957}$$

$$\boxed{P \Rightarrow 140 \text{ N}}$$

Q Determine the horizontal force, "P" needed to just start moving the 300N crate up the plane. $\mu_s = 0.3$.

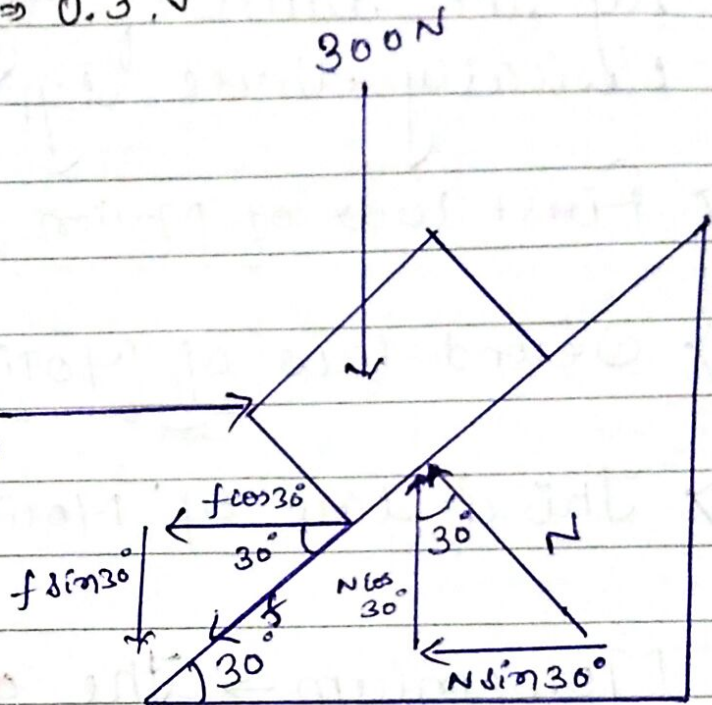
Soln

$$\sum F_x = 0$$

$\pm \rightarrow$

$$P - f \cos 30^\circ - N \sin 30^\circ = 0$$

$$P = f \cos 30^\circ + N \sin 30^\circ \quad (1)$$



$$\sum F_y = 0$$

$$N \cos 30^\circ - f \sin 30^\circ - 300 = 0$$

$$N \cos 30^\circ - 0.3 N \sin 30^\circ - 300 = 0$$

$$f = \mu N$$

$$f = 0.3 N$$

$$N [0.866 - 0.3 \times 0.5] = 300$$

$$N_{\text{normal}} = 419.7$$

$$f = 0.3 \times N$$

$$= 0.3 \times 419$$

$$f = 125.7$$

$$P = f \cos 30^\circ + N \sin 30^\circ$$

$$P = 125.7 \cos 30^\circ + 419 \sin 30^\circ = 318.35$$